

BELL: A Fully-Collateralised, No-Liquidation Market for Closed-Session Equity Gap Risk

Adarsh Ron

Independent researcher

September 2026

PROTOCOL DESIGN · EMPIRICAL CALIBRATION · ON-CHAIN VERIFICATION

Abstract. A venue that trades continuously while the market it references trades for 19.4% of wall-clock time manufactures a recurring, measurable, and currently unhedgeable jump risk. This paper designs, prices, calibrates and verifies an instrument that monetises it. **BELL** issues a pair of complementary ERC-20 claims — GAP-L and GAP-S — on the closed-session gap $G = O/C - 1$ of a tokenised equity. One unit of collateral mints exactly one unit of each claim, so the two always sum to one. This makes full collateralisation a *theorem* rather than a risk-management practice: there is no margin requirement, no maintenance threshold, no liquidation engine, and no realised pair of prices at which the protocol owes more than it holds. The payoff is a capped absolute-move claim, $\Pi_L = \min(\lambda|G|, 1)$, which bounds liability at the collateral unit while preserving the hedge's linearity across the economically relevant range. Leverage is *derived* from a stated saturation probability rather than chosen from a chart, $\lambda^* = 1/Q_{1-\alpha}(M)$.

The paper's principal contribution beyond the design is measurement. Using 53,112 close-to-open gaps for 22 US large caps over 2014–2026, I show that the illustrative volatilities a session-dated product is normally calibrated on are wrong in both directions: overnight volatility is **understated by 36%** for the high-beta names the product targets, while weekend volatility is **overstated by 13%**. The weekend variance ratio is 0.356, not the 0.88 a three-increment model implies. Corrected inputs invert the design's own leverage ranking of the two sessions, and the Gaussian pricing formula overstates the fair premium by **+29 to +42% overnight** against the empirical distribution, a defect a variance-matched Student-t only partially repairs and a fitted normal-inverse-Gaussian closes to within 2%. A second contribution is a settlement-integrity result: the intuitive fallback rule, which refunds both legs at half value when no valid print arrives, is a **free long butterfly struck at half the cap**, written by the protocol at no premium to its own hedgers, worth 29.7 basis points of notional per session and 14.3% of the premium paid. Deferred settlement removes it exactly at a cost of 0.021 bp.

The design is verified by 105 executable numerical checks (94 pass, 0 fail) and by an end-to-end deployment on a fork of Robinhood Chain at chain ID 4663, where the sum-to-one invariant holds bit-exactly, the settlement lifecycle drains collateral to zero, and twelve documented failure paths resolve without breaking solvency. I close with the viability constraint that the analysis exposes: revenue is a function of *open interest held over time*, and the inventory required to reach break-even is a large multiple of the tokenised-equity base that currently exists.

Keywords: closed-session gap risk · complementary claims · full collateralisation · constant-product market making · implied volatility · tokenised equities · settlement integrity · outcome-token AMM

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1. Introduction

On 1 July 2026 Robinhood Chain opened its mainnet: a permissionless, EVM-compatible Ethereum Layer 2 built on the Arbitrum Orbit stack, with chain ID 4663, ETH as the gas token, roughly 100 ms block times and native ERC-4337 account abstraction. Its stated purpose is to host tokenised equities — *Stock Tokens*, tokenised debt securities issued by a Jersey vehicle that provide economic exposure to a reference share without conferring ownership, voting rights, or the corporate-law treatment of the underlying.

These instruments are tradable 24/7. The reference market they track is not. The US equity regular session runs 09:30–16:00 ET, five days a week: 32.5 hours out of 168, or **19.35%** of wall-clock time. A Stock Token is therefore transferable, collateralisable and DeFi-composable for **80.65%** of its life while referenced to a price that is not updating. The issuer's own documentation states the consequence plainly: 24/7 trading does not equal 24/7 pricing of the underlying, and when the reference market is closed the on-chain price depends on the last traded price, market-maker models, oracle inputs and participant expectations.

This asymmetry is the chain's defining structural feature, and it manufactures four distinct risk vectors — the session gap itself, the discovery error in the 24/7 price, collateral liquidation at a stale oracle, and event front-running — of which the first is both the largest and the least hedgeable. The window in which gap risk crystallises is precisely the window in which listed options are shut.

THE STRUCTURAL CLAIM

Only an instrument that settles against the *official external print* — a number produced off-chain, by the primary listing venue, at a timestamp the chain does not control — can hedge the closed-session gap. And that instrument must be tradeable *during* the closed session, which means it must be on-chain. Satisfying both conditions simultaneously is what makes the instrument possible at all, and it is the property that separates it from every existing alternative.

1.1 Contributions

1. **A design for the instrument.** A pair of complementary ERC-20 claims on the absolute session gap, fully collateralised by construction, market-made by a constant-product AMM whose marginal prices sum to one identically, with a term-proportional protocol fee and a trading fee that rises into the open to compensate liquidity providers for adverse selection (§4, §6).
2. **A solvency theorem, not a risk practice.** The liability equals the collateral for every realised pair of prints and every leverage setting — verified at a residual of exactly zero over 484,121 grid points and 400,000 randomised draws (§4.2, §9.1).
3. **A pricing framework and a byproduct.** A closed-form truncated absolute moment gives the fair premium, which inverts uniquely into an implied session volatility. The protocol publishes a continuous, permissionless implied-volatility term structure for closed-session risk on every listed token — a risk input for lending collateral parameters and market-maker quoting, independent of the derivative itself (§5).
4. **Measurement that corrects the design's own inputs.** 53,112 measured close-to-open gaps show the overnight volatility assumption understated by 36% and the weekend assumption overstated by 13%, the weekend variance ratio at 0.356 rather than 0.88, the Gaussian pricing formula overstating the premium by 29 to 42% overnight at the per-name leverage, and a fitted normal-inverse-Gaussian closing that error to within 2% (§7).

5. **A settlement-integrity result.** Five routes into the reference process are enumerated; the design's manipulation-immunity claim is proved and sharpened to cover one of them; and the intuitive fallback is shown to be a free butterfly written to the protocol's own hedgers, with a quantified repair (§8).
6. **End-to-end verification.** 105 executable numerical checks and a deployment on a fork of the live chain, including twelve documented failure paths, with the findings that survive (§9).

1.2 Scope

This is a protocol design and empirical-calibration paper. It is not investment, legal, tax or accounting advice. The distinction between measured data and illustrative assumption is marked explicitly throughout; §7 and §11 are measurement, §4–§6 are design, and §8 is analysis. Every figure reported as measured is traceable to a cached dataset or an on-chain read, and every figure reported as illustrative is labelled as such.

2. Related work

2.1 Complementary claims and the sum-to-one primitive

The construction used here belongs to a family of mechanisms in which a bounded payoff space is partitioned into two claims whose values sum to one, rather than matched against a counterparty. The distinguishing property is that a single unit of collateral mints exactly one unit of each leg, so the system is fully funded at every point rather than on average. This removes the two costs that make on-chain options expensive — counterparty matching and collateral lockup — at the price of bounding the payoff.

Prior deployments of the primitive have been calibrated on monthly terms against Ethereum mainnet positions. That calibration carries three parameter choices which do not transfer to short-dated instruments, and naming them precisely frames the contributions of §6:

- **A flat protocol fee.** A fee specified as a fixed percentage of collateral on redemption is not one number but a family of annualised rates: the same charge is 12.17% annualised on a monthly position, 133.7% over a weekend, and 500.6% on an overnight session — a spread of $41.1\times$ across products (§6.3, verified at K10 and P11). A sub-day instrument requires the fee to be specified as a *rate*.
- **Leverage chosen by inspection.** Selecting the leverage factor from historical charts gives no rule for a new index, no response to a volatility regime change, and no stated probability of saturation. §6.1 replaces it with a quantile rule.
- **Monthly terms.** Where settlement is measured in weeks, the economics of the instrument change; the entire product becomes a function of a session calendar that did not previously exist.

2.2 Market making for outcome tokens

A constant-product AMM is a poor fit for a claim that terminates at a known time, and the literature says so explicitly. In work on uniform AMMs for prediction markets, Moallemi and Robinson observe that outcome-token volatility depends on both the current probability and the time to expiry, so the pool provides inconsistent liquidity; they note that liquidity providers are "essentially guaranteed to lose all of their value once the prediction market expires," and formalise the loss as *loss-versus-rebalancing* (LVR) — the rate at which an AMM loses money to arbitrage, which depends jointly on the AMM's shape and the price process of the asset traded on it. Their derivation produces an invariant specifically for complementary outcome-token reserves.

This is the single most important external result bearing on the design, for two reasons. First, it independently corroborates the LP break-even problem derived from first principles in §6.4 — that LPs in this product class are structurally short the underlying move and must be compensated for it. Second, the separate literature on replicating market makers establishes the converse direction: every payoff function that is concave, nonnegative, nondecreasing and 1-homogeneous has a corresponding convex CFMM, with a constructive method for recovering the trading function. BELL's AMM is therefore an application of a known equivalence rather than an open problem.

The practical consequence, developed in §6.4 and §11, is that the binding constraint on this product is *not* trader demand. It is whether the liquidity-providing side can be compensated sufficiently for a position that is short volatility by construction. That reframing drives the economic analysis.

2.3 On-chain options and structured products

Three live protocol families occupy adjacent ground. Perpetual options protocols that reinterpret concentrated-liquidity positions as written options provide downside structure and additional LP yield without an oracle. Yield-tokenisation protocols split a yield-bearing asset into a principal claim and a yield claim with fixed expiries, and run a specialised AMM designed for time-decaying assets; this is structurally the closest live implementation of a claim pair with a purpose-built market maker, and it demonstrates that the mechanism class operates at scale across multiple chains. European-options AMMs provide the third reference point. None of them, however, settles against an off-chain print: their settlement is a function of prices their own venues produce, which is precisely the property BELL is designed to avoid (§8).

2.4 Tokenised equities

The infrastructure BELL requires exists and shipped in 2026. On-chain, permissionless tokenised stocks and ETFs with reserve attestation are live and integrated with lending markets; commercially, tokenised US equity trading launched in mid-2026 with 24/7 trading following within weeks. The relevant consequence for this paper is that the tokenised-equity base is real but early, which sets the scale constraint analysed in §11.

3. The closed-session gap

3.1 Definition

Fix a canonical Stock Token X and a session k , defined as the interval between a reference-market close at time t_k^- and the following open at t_k^+ . Let C_k be the official closing print and O_k the official opening print, both sourced externally and timestamped by the producing venue. The *session gap return* is

$$G_k = O_k / C_k - 1 \quad (1)$$

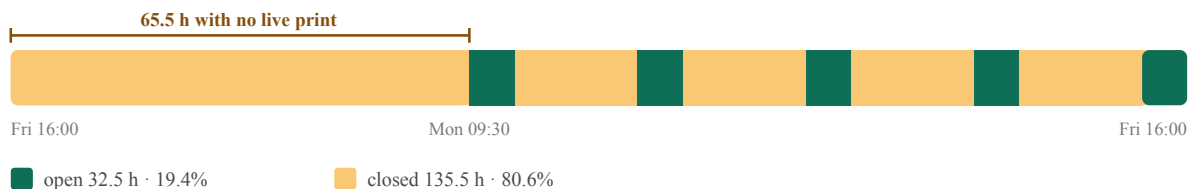
The signed gap is directional; its magnitude is a volatility quantity. The flagship instrument settles on the magnitude, because the dominant demand is from holders of inventory who are short the gap in *both* directions. An important implementation point, established in §7.6: settlement must be on the *multiplier-adjusted token price*, not the headline share price. Because the multiplier reinvests dividends, a share price drops by the dividend on an ex-date while the multiplier rises to offset it; a protocol settling on the raw share price would record a spurious gap on every ex-date, roughly quarterly per name.

3.2 The 19.4% arithmetic

The US equity regular session spans 6.5 hours per day, 32.5 hours per week, out of 168 hours in a week:

$$32.5 \text{ h} / 168 \text{ h} = 19.35\% \quad (2)$$

A token referenced to that market therefore trades without a live print for 80.65% of the time it is transferable. The weekend block alone spans 65.5 consecutive hours, and the holiday blocks extend to 89.5 hours. Across a year this produces roughly 252 session events per name, each of which is a discrete, unpriced transfer of value between holders of the token.



The window in which gap risk crystallises is exactly the window in which listed options are shut.

Figure 1. Reference-market coverage of one wall-clock week, drawn from Friday 16:00 ET. Green denotes the regular session, during which the reference market produces a live print; amber denotes a closed session, during which the token remains transferable but the reference price is stale. The gap the protocol settles on is the discontinuity across each amber band.

3.3 Four risk vectors

Table 1. Risk vectors created by the 24/7-versus-closed-market mismatch, and whether BELL addresses them.

Vector	Mechanism	Frequency	BELL response
Session gap	The official open differs from the prior close; the holder is unhedged across the discontinuity.	~252 / name / yr	Primary instrument (GAP)
Discovery error	The 24/7 token price is a forecast of the open and can be wrong. Anything pricing risk off it inherits the error.	continuous	Extension (CAL, \$10.2)
Collateral liquidation	A lending market holding the token as collateral can be liquidated by a gap at a moment when the oracle is stale.	~252 / name / yr	Primary demand source (\$11.1)
Event front-running	News, halts and earnings hit the token market first and are corrected later; the correction is the gap.	name-specific	Event-session term (\$6.2)

3.4 Why existing instruments cannot hedge this

Table 2. Candidate hedges for the closed-session gap and the specific reason each fails.

Instrument	Why it cannot hedge the closed-session gap
Listed single-stock options	Traded only during the regular session — the exact window in which the risk does not exist — and expiring on a monthly cycle, so a Friday-close-to-Monday-open exposure cannot be isolated. No listed options exist on a tokenised debt instrument, and a non-US retail base has no access to US options in any case.
Perpetuals on the token	Priced off the same 24/7 token price series. Hedging a gap with an instrument that <i>is</i> the gap is circular, and perps import funding, margin and liquidation risk into a hedge whose purpose is to reduce risk.
On-chain volatility products	Settle against chain prices, so settlement is manipulable by whoever can move a thin pool at the observation timestamp.
Holding cash / reducing size	Not a hedge but a liquidation: it forgoes the desired exposure and pays transaction costs twice.

4. Protocol design

4.1 Architecture

Three modules and one invariant. A **settlement engine** consumes only external prints for the GAP instrument, which is what makes on-chain price manipulation irrelevant to settlement. A **mint-and-redeem module** enforces the sum-to-one relation between collateral and the two claims. A **constant-product AMM** provides continuous pricing without an external venue. A fee engine applies a rising trading fee and a capped protocol fee.

4.2 The sum-to-one invariant and the solvency theorem

Depositing one unit of collateral mints exactly one Long token and one Short token, and the reverse holds less a protocol fee:

$$1 \text{ U} \rightarrow 1 \text{ GAP-L} + 1 \text{ GAP-S} \quad 1 \text{ GAP-L} + 1 \text{ GAP-S} \rightarrow 1 \text{ U (less } \eta)$$
 (3)

Because the two are complementary claims on one unit of collateral, $\Pi_S = 1 - \Pi_L$ identically.

Theorem 1 (Solvency). *Suppose the settlement contract has minted N pairs against N units of collateral. Then for any realised gap G and any $\lambda > 0$, the total settlement liability is*

$$N \cdot [\min(\lambda|G|, 1) + (1 - \min(\lambda|G|, 1))] = N$$
 (4)

which equals the collateral held, for every G .

Therefore: no margin requirement, no maintenance threshold, no liquidation engine, no path dependence, and no scenario in which the protocol owes more than it holds. The consequence deserves emphasis. There is no price at which a BELL position can be force-closed, because there is no state in which the collateral fails to cover the claim. A trader's maximum loss on a long GAP-L position is the premium paid; the maximum loss on the short leg is the collateral posted. In a market whose participants are largely retail, holding positions across a 65-hour weekend with no ability to monitor a margin call, this is not a convenience feature — it is the difference between a product that can be offered and one that cannot.

Note carefully what Theorem 1 does *not* say. It is unconditional in the prints (C, O) , and therefore says nothing about them. Settlement factors as

$$\text{world state} \rightarrow (C_k, O_k) \rightarrow G_k \rightarrow \Pi_L$$
 (5)

and the theorem covers only the last two arrows. The first — the map from the world to the pair of prints — is the entire residual risk of the protocol, and it is not a mathematical object but a procedure. §8 is devoted to it.

4.3 Payoff specification

The flagship instrument settles on the magnitude of the gap, with a cap at one collateral unit and a leverage factor λ_k that narrows the domain:

$$\Pi_L^k = \min(\lambda_k \cdot |G_k|, 1) \quad \Pi_S^k = 1 - \Pi_L^k$$
 (6)

Read economically, Π_L is a *capped straddle*: it pays linearly in the absolute size of the gap up to a saturation point at $|G| = 1/\lambda$, then flat. Π_S is the funded short-volatility leg. The cap does something a vanilla straddle cannot: it bounds the payoff above by the collateral unit, which is precisely what allows full collateralisation with

no margin machinery. The payoff is an even function of the signed gap, verified at a residual of exactly 0.0e+00 over $\pm 50\%$ at every session type (check S1) — so the *direction* of any induced move carries no value, only its magnitude.

4.4 The constant-product AMM and the exact cost function

With a GAP-L and b GAP-S in the pool, the invariant $a \cdot b = \kappa$ gives marginal prices

$$p_L = b / (a + b) \quad p_S = a / (a + b) \quad \rightarrow \quad p_L + p_S = 1 \quad (7)$$

The pool can never quote a set of prices that admits a parity arbitrage. Critically, there is no collateral/Long pair: a buyer does not swap collateral for Long. She deposits x collateral, the contract mints x Long and x Short, and she swaps her Short into the pool for additional Long. The collateral never leaves the settlement contract, which keeps the invariant *exact* rather than approximate.

To acquire Q Long tokens the pool moves from (a, b) to (a - y, b + x) with $x + y = Q$, and the constant-product constraint gives a quadratic in x:

$$x^*(Q) = [-(a+b-Q) + \sqrt{(a+b-Q)^2 + 4Qb}] / 2 \quad (8)$$

Equation (8) is the *rationalised* form, and the choice matters numerically. The textbook quadratic formula computes the same quantity as a difference of two nearly equal numbers and loses all significance as the trade size falls relative to pool depth: at $Q/n \sim 1.0e-16$ the naive form has a maximum relative error of 2.88 — i.e. no correct digits at all — while the rationalised form remains exact (check M10). On-chain, a trade of one base unit against a 500,000-unit pool returns 500000000000500000 wei under the rationalised form and 500000000000000000 wei under the naive one (suite E, check E2) — the rewrite buys roughly five orders of magnitude in a context where the error is the entire signal.

For validation against published reference cases, equation (8) reproduces the canonical worked example exactly: with $a = 1,000$, $b = 200$ and $Q = 20$, it returns $x^* = 3.3801$ against a published 3.38, an average price of 0.1690 against a marginal price of 0.167 (checks M7, K8).

4.5 Slippage law and the depth requirement

Expanding equation (8) to second order in trade size relative to depth gives, with $n = a + b$:

$$x^*(Q)/Q - p_L \approx (Q/n) \cdot p_L(1-p_L) \quad (9)$$

Slippage is linear in trade size and inverse in depth. Since $p_L(1-p_L) \leq 1/4$, a sufficient depth condition for a trade of size Q at slippage no worse than s is

$$n \geq Q \cdot p_L(1-p_L) / s \quad (10)$$

The residual of the first-order law decays as $(Q/n)^2$ and the third-order correction as $(Q/n)^3$, confirming the expansion order (check M8). One observation from the differential suite is worth recording as a design property rather than a defect: at $p_L = 1/2$ the third-order correction factor $1 - e(2p - 1)$ is identically 1, so the higher-order term does no work at the price at which a session opens by construction. The correction is real but inert at the design point (check F8, suite C6).

4.6 No-arbitrage band

Selling a minted pair back to the contract yields $1 - \eta$ units, so if $p_L + p_S$ fell below that level a trader would buy both legs and redeem; if the sum exceeded one, a trader would mint a pair for one unit and sell both legs. Hence

$$1 - \eta \leq p_L + p_S \leq 1 \quad (11)$$

Because the AMM prices satisfy the sum-to-one relation identically, the band is tight at its upper edge — measured at a deviation of at most $2.2\text{e-}16$ across 20,000 pools (check K11) — so the protocol is arbitrage-consistent by construction rather than by monitoring.

5. Pricing and implied volatility

5.1 The truncated absolute moment

Model the gap as approximately Gaussian, $G \sim N(0, \sigma^2)$, with the fat-tail correction discussed below. With $c = 1/\lambda$ the saturation point and $Z = G/\sigma$ standard normal, the expected payoff per unit of leverage is the truncated first absolute moment:

$$E[\min(|G|, c)] = 2\sigma \cdot (\varphi(0) - \varphi(c/\sigma)) + 2c \cdot (1 - \Phi(c/\sigma)) \quad (12)$$

Derivation. Split the expectation at $|G| = c$. The inner integral is $\int u \varphi(u) du = -\varphi(u)$; the outer term is $c \cdot P(|G| \geq c)$. $\square$

The fair value of the Long token is then $p_L = \lambda \cdot E[\min(|G|, c)]$. Two limits confirm the expression: as $c \rightarrow \infty$ the second term vanishes and the first tends to $\sigma\sqrt{2/\pi}$, the expected absolute move, as it must; as $c \rightarrow 0$ the payoff saturates immediately and $\lambda E[\cdot] \rightarrow 1$. Both limits are verified, along with agreement against quadrature at $8.99\text{e-}16$ across 35 parameter combinations and against 4,000,000-path Monte Carlo within five standard errors (checks M11–M13).

5.2 BELL-IV: a volatility oracle by construction

Because $E[\min(|G|, c)]$ is strictly increasing in σ and maps $[0, \infty)$ onto $[0, c)$ — the minimum discrete derivative over 5 truncation ratios $\times$ 400 volatilities is $4.023\text{e-}01 > 0$ (check M14) — the market price p_L inverts uniquely into an implied session volatility:

$$\hat{\sigma}_k = \text{the unique } \sigma \text{ solving } \lambda_k E[\min(|G|, c)] = p_L \quad (13)$$

The inversion round-trips to $1.16\text{e-}13$ relative across 20 combinations (check M15), and its elasticity is unity to within 0.21% while $c/\sigma \geq 3$, so a 1% relative price error produces a 1% relative volatility error (check X8). The protocol therefore publishes, continuously and permissionlessly, an implied-volatility term structure for closed-session risk on every listed token. This has value independent of the derivative itself: it is a risk input for lending-protocol collateral parameters, for market-maker quoting, and for any agent that must price a tokenised equity over a weekend. One honest limitation applies and is handled as a design guard rather than a caveat: BELL-IV is a function of the pool price, so a pool that has not traded reports the old volatility exactly. The fix is provenance stamping with a freshness bound and a named trailing-realised fallback (check G2/T2).

5.3 Model risk, stated honestly

Equity close-to-open returns are strongly leptokurtic, and §7.4 measures the consequences directly. Two properties of the design contain the damage. First, leverage is set from an empirical quantile, never from a Gaussian formula (§6.1). Second, the cap at c means tail misestimation costs the holder only unhedged residual exposure beyond the cap: it cannot produce a loss larger than the collateral and cannot make the protocol insolvent. The structure is chosen so that model error is a *performance* question rather than a *solvency* question. A comparison of the lognormal and Gaussian truncated moments at equal variance shows the two differ by at most 0.18% (check M17), so the choice of lognormal-versus-normal is second-order relative to the tail question in §7.4.

The launch pricing model. Because the distribution is the thing that is wrong (§7.4), the protocol does not launch on a parametric form. The primary model is the *empirical truncated distribution*: the truncated absolute moment of equation (12) evaluated directly on the per-name, per-session sample of §7.1, which is the estimator that is unbiased for a pricing question by construction. Two qualifications define its use. First, the cap sits at the 99th percentile while the overnight sample is roughly 190 observations per name per year, so the truncation point lies in the extreme tail of the sample and the empirical estimate of the premium is dominated by the body, where the data are dense. That is the right property for a premium and the wrong one for a risk limit, which is why leverage is set from a quantile with a model-based tail extension rather than from the same sample (§6.1). Second, where a name's sample is too thin to place the truncation point at all, a fitted fat-tailed alternative is used and the fit is published as part of the parameter set: a normal-inverse-Gaussian, or a jump-diffusion with separately estimated jump intensity and jump size, both of which admit the truncated moment in closed or quadrature form. A variance-matched Student-t is rejected as a candidate on measurement rather than on principle — at $v = 4$ it leaves a +30% overnight residual against the empirical distribution — and it is the only alternative the design had previously named.

Calibration is per name and per session type, never class-level. The measured overnight cross-section runs from 1.09% for AAPL to 3.13% for COIN (§7.2), so a single class number used as a seed ratio would be wrong in both directions on the same day. The seed ratio published at launch is therefore the table of §7.8, indexed by name and session, and every entry records the model that produced it. This also makes the pricing error of §7.4 auditable after the fact rather than merely disclosed before it: each session's realised gap can be scored against the model that seeded its pool.

Both fitted families are fitted, priced and costed in §7.11. The short version is stated here so that the design section does not leave its own open question open: the fitted fat tail closes the Gaussian pricing error almost completely, the fat-tailed family the design named does not, and evaluating the fit on-chain costs two orders of magnitude more than publishing its output — so the fit belongs off-chain.

6. Parameterisation

6.1 Leverage, derived rather than backtested

Let $M = |G|$. The designer faces a two-objective problem: maximise capital efficiency, which means maximise λ ; and bound the probability that the claim saturates, which means keep $c = 1/\lambda$ large. Requiring the saturation probability to be at most α and taking the largest admissible λ gives the rule

$$\lambda^* = 1 / Q_{1-\alpha}(M) \quad (14)$$

where $Q_{1-\alpha}(M)$ is the $(1-\alpha)$ quantile of the absolute session gap, estimated from realised history for that name and that session type, over a trailing window of 252 to 504 sessions re-estimated every session (§7.9). Under a Gaussian approximation this reduces to $\lambda^* = z_{1-\alpha/2}^{-1}/\sigma$ — leverage inversely proportional to session volatility, which is the qualitative rule the earlier literature reached empirically. The derivation replaces that intuition with a stated probability and a stated estimator, which makes the parameter governable, auditable, and responsive to regime change.

The rule's lattice sensitivity is a real design constraint that is easy to miss. The published leverage values depend on the rounding lattice, which the design never states: a 1% lattice gives $\lambda_E = 16$, a 0.5% lattice gives 18, and 0.25% gives the published 19. The lattice is therefore load-bearing and must be specified (checks S4, T8).

Property 1 (Rule validity). *Fed measured inputs, the rule delivers the property it promises: realised $P(\text{saturate}) = 1.00\%$ overnight and 1.02% on weekends against a stated $\alpha = 1\%$ (check P4). Fed the design's illustrative inputs it does not: at $\lambda = 19$ the overnight session saturates on 2.74% of sessions, $2.74\times$ the advertised rate. The derivation is correct; only its inputs were wrong.*

6.2 Session taxonomy

The natural clock for this product is the market session calendar, and different session types have genuinely different risk characteristics. This produces a term structure where the earlier literature had a point.

Table 3. Session taxonomy. The design column lists the illustrative magnitudes the protocol was originally specified against; §7 replaces them with measured values.

Code	Session	Window	Duration	σ (design)	σ (measured, high-beta)	Freq / yr
E	Overnight	Mon–Thu 16:00 → next 09:30 ET	17.5 h	1.60%	2.18%	~191
W	Weekend	Fri 16:00 → Mon 09:30 ET	65.5 h	2.60%	2.25%	~52
H	Holiday	Pre-holiday close → next open	17.5–89.5 h	2.00%	1.82%	~9
C	Event	Session containing a scheduled release	as defined	$\gg \sigma_E$	name-specific	—

Three consequences follow. Each session type gets its own λ , since $\lambda^* = 1/Q_{1-\alpha}(M)$ and the distributions differ; the protocol offers a menu, not a single parameter. Event sessions must be isolated, because a session containing a scheduled earnings release has a materially different distribution and including it in the overnight pool would corrupt the calibration and systematically under-price the risk. And the annualisation must respect the taxonomy rather than naively scaling one session:

$$\sigma_{\text{ann}} = \sqrt{N_E \sigma_E^2 + N_W \sigma_W^2 + N_H \sigma_H^2} \quad (15)$$

with $N_E + N_W + N_H \approx 252$. On the design's illustrative values this gives $\sigma_{\text{ann}} = 29.61\%$ against 25.4% from a naive $\sigma_{\text{session}} \sqrt{252}$, with weekend sessions carrying 40.1% of total variance while occupying 39% of wall-clock time (checks P13, K5).

The weekend variance ratio isolates a new factor. If gap increments were independent and identically distributed, weekend variance would be three times overnight variance; empirically it is not, because weekend information arrival is lower.

$$\text{WVR} = \sigma_W^2 / (3\sigma_E^2) \quad (16)$$

A trader can go long the weekend instrument and short three overnight instruments, isolating WVR as a pure factor with no directional equity exposure and no dependence on the level of volatility. This is a risk factor that exists only because the chain runs 24/7 against a market that does not. §7.3 measures it, and the measurement is materially different from the design's assumption.

6.3 Fee design

Two fees exist: a trading fee paid to liquidity providers, and a protocol fee on redemption. The protocol fee must be specified as an annualised rate prorated by session length, not as a flat amount:

$$\eta_k = \eta_{\text{ann}} \cdot T_k / (365 \times 24 \text{ h}) \quad \eta_{\text{ann}} = 12\% \quad (17)$$

Table 4. Protocol fee by session type under the annualised specification.

Product	Duration	η_k	Basis points
Overnight session E	17.5 h	0.0240%	2.40 bp
Weekend session W	65.5 h	0.0897%	8.97 bp
Holiday session H	up to 89.5 h	0.123%	12.26 bp
<i>Flat-fee design, for reference</i>	30 d	1.00%	100 bp

The defect being repaired is not the magnitude of a flat fee but that a flat fee prices time at a rate that varies by a factor of forty depending on which product it is attached to. Measured: the same flat 1% annualises to 500.6% on an overnight session, 133.7% over a weekend and 12.17% over a month — a spread of $41.1 \times$ (checks K10, P11). Under equation (17) the implied annual rate is 12.00% on every term by construction (check P10).

A second, independent constraint guards against the case where the premium compresses without the term changing:

$$\eta_k \leq \beta \cdot p_L^{(k)} \quad \beta \approx 0.05 \quad (18)$$

At a premium of 0.228 the guardrail permits 114 bp against a 9.0 bp charge, so it does not bind in normal conditions and binds only under stress — which is what a backstop is for (check P12). Together the two rules give a fee that is economically constant across terms in normal conditions and self-limiting in abnormal ones, with no governance intervention required in either case.

The trading fee should rise into the open, because adverse selection does. As t approaches the open the gap is progressively revealed through pre-market signals, foreign-listing cross-reads, index futures and single-name news, so informed flow arrives late and liquidity providers face their maximum adverse selection in the final hours:

$$\varphi(t) = \varphi_0 + (\varphi_1 - \varphi_0) \cdot (t - t_k^-) / T_k \quad \varphi_0 = 0.10\%, \varphi_1 = 1.00\% \quad (19)$$

Equation (19) ramps the fee through the session. It does not scale the fee *level* with volatility, and the two are not equivalent, because the time-average of a ramp bounded by φ_0 and φ_1 is at most $\varphi_0 + (\varphi_1 - \varphi_0) \cdot 2/3 = 0.70\%$ whatever the intraday shape. That ceiling is the binding fact about the tolerance range stated in §6.4. The low end of the range — 3.79% of pool value at $1 \times$ turnover — is reproduced exactly by the time-average fee of 0.55%, and the premium it implies, 0.1451, sits inside the measured range of §7.8. The high end — 55.11% at

10× turnover — requires a fee of 0.7997%, which is 1.45× the time-average and above the 0.70% ceiling. It is not reachable by reweighting the ramp, which caps out at 48.24%.

The break-even condition says why, and gives the form the fee has to take. Writing $t_{\max}(\phi, \tau) = \phi \cdot \tau / (w \cdot p_L)$ for the net imbalance an LP can tolerate, the required ϕ is linear in the premium, and the premium is linear in σ over the range that matters: measured across $\sigma = 1\text{--}3\%$, the ratio p_L/σ is constant to within 1.2%. A fee that must cover a liability proportional to σ has to be proportional to σ as well, and the design should say so rather than leave it to be inferred:

$$\phi(\sigma) = \phi_{\text{ref}} \cdot (\sigma_{\text{realised}}/\sigma_{\text{ref}}), \text{ capped at } \phi_{\text{max}} \quad \phi_{\text{ref}} = 0.55\% \text{ at the calibration reference} \quad (20)$$

The three candidate schedules are simulated below on the measured per-name volatilities and premiums, with turnover drawn from a lognormal distribution calibrated to a median of 2× and a 90th percentile of 10×.

Table 5. Trading fee under the three candidate schedules. Time is equation (19); reweight redistributes the same ramp toward high-volatility moments; level scales the fee by realised σ as in equation (20). Each fee column is the median of the simulated distribution. The last column is the share of draws whose fee covers the LP's short exposure at the low anchor.

Name	Session	σ	p_L	Time	Reweight	Level	P(fee > 3.79%)
NVDA	E	1.88%	0.1740	6.29%	8.01%	6.29%	65.8%
NVDA	W	2.15%	0.1413	7.75%	9.86%	7.75%	71.7%
NVDA	H	1.62%	0.2555	4.29%	5.46%	4.29%	54.0%
TSLA	E	2.18%	0.1465	7.47%	9.51%	7.47%	70.7%
TSLA	W	2.57%	0.1693	6.47%	8.23%	6.47%	66.6%
TSLA	H	2.72%	0.1799	6.09%	7.75%	6.09%	64.8%
AAPL	E	1.09%	0.1440	7.60%	9.68%	7.61%	71.2%
AAPL	W	1.45%	0.1306	8.38%	10.67%	8.38%	73.7%
AAPL	H	1.12%	0.2356	4.65%	5.92%	4.65%	56.5%

Two things in the simulation are stated because they cut against the tidy version of the argument. The first is that level scaling does not raise the median: NVDA's overnight median is 6.29% under both the time and the level schedule, because the median turnover is near 1× and the level multiplier is near 1 there. What equation (20) changes is the mean and the upper tail — the 90th percentile moves from 31.5% to 32.0% and the 10th from 1.26% to 1.24% — which is where the LP's exposure actually is, but it is a redistribution toward the tail rather than a uniform increase, and presenting it as the latter would be wrong. The second is that the low anchor is the robust one: 3.79% is reproduced by measured parameters with no free choice, whereas the high anchor depends on which session it is attached to.

That ambiguity should be resolved in the text rather than left to the reader, because the two readings imply different fee mechanisms and only one of them the ramp can deliver. At the reweighting ceiling the high anchor is attainable only for a session whose premium is about 0.1270; AAPL's measured weekend premium is 0.1306, so on the evidence 55.11% describes a low-volatility *weekend* session, not an overnight one. Read as an overnight figure it is unreachable without equation (20); read as a weekend figure it is reachable at the ceiling. The design should say which it means.

6.4 LP economics

Liquidity providers are the sellers of the options, and their economics must close. An LP deposits x collateral, receives x Long and x Short, deposits them into the pool at the pool's prevailing ratio — a price-neutral deposit rule, which prevents the LP's own deposit from moving the price and creating an arbitrage, verified at 0e-12 relative price movement on both add and remove (suite C, check C7) — and retains the unmatched leg.

At settlement, since $\Pi_L + \Pi_S = 1$, the LP's pool share w of final balances (a' , b') is worth

$$w [a'\Pi_L + b'\Pi_S] = w [b' + (a' - b')\Pi_L] \quad (21)$$

If traders have been net buyers of Long then $a' < b'$ and the LP's payoff is *decreasing* in the realised payoff: the LP is short the session move. Measured, an LP's share value falls monotonically from 1,161,187 to 861,187 as the payoff runs $0 \rightarrow 1$, a total swing of 300,000 (check P18). This yields a break-even condition on fee revenue:

$$\varphi \cdot V \geq w \cdot |a' - b'| \cdot E[\Pi_L] \quad (22)$$

where V is session trading volume. The right-hand side is the LP's expected short exposure, which the fee schedule of equation (19) must cover. Note the structural difference from a conventional option market: an LP here is not writing naked options with unbounded liability. The pool holds only tokens fully backed by collateral the LP itself posted, and the worst case is that the retained leg expires worthless — a loss bounded by the collateral, with no margin call at any point.

This is the point at which the external literature matters most. LPs in this product class are short volatility by construction, and the formal treatment of outcome-token AMMs identifies loss-versus-rebalancing as the mechanism by which they lose. §11.3 quantifies the consequence: the required subsidy scales with open interest, and above a modest net-flow imbalance it exceeds the protocol's entire revenue base. The binding constraint on this product is the LP side, not the trader side.

The closure condition. Equation (22) has to hold at launch volume, not at scale, and it does not hold unconditionally. Solving it for the net-flow imbalance an LP can absorb gives roughly 3.79% of pool value at $1\times$ turnover under time-average fees, rising to 55.11% at $10\times$ turnover under volatility-weighted fees (§11.3). The binding case is the low-turnover one: at $1\times$ turnover the pool closes only while a session's net Long flow stays inside $\pm 3.79\%$. Above that, the fee schedule no longer covers the short exposure the LP has written and the LP is being paid less than the option it sold. Two facts make this a launch requirement rather than a tuning exercise. First, the required subsidy above the threshold scales with open interest rather than with revenue: an 8% imbalance implies about \$13.7M a year, roughly 91% of open interest, and 15% implies \$52.1M, or 347% (§11.3). Second, the distribution of net Long flow is a property of a market that does not yet exist, so the condition cannot be verified in advance (§12.1). It can only be measured, and the design must therefore be launchable in a state where the measurement is cheap and the remedy is bounded.

A cold-start programme. Four levers bring the pool inside the closure condition, in increasing order of cost:

1. **A capped, curated name list.** Listing a handful of the deepest, highest-beta names rather than the full universe concentrates both hedging demand and speculative flow into pools that can be watched individually, and it makes the imbalance measurement meaningful: a 3.79% tolerance on a \$0.5M pool is a \$19,000 imbalance, which is observable in weeks rather than quarters. The list is a launch parameter, not a permanent constraint.

2. **Inventory-backed seeding.** A market maker that commits inventory to both legs at inception supplies the balancing flow that hedging demand consumes, and is compensated through the fee schedule rather than by a protocol transfer. This is the only lever that produces imbalance tolerance without spending protocol capital, and it is why the first lever matters: a committed maker will only underwrite a list short enough to price.
3. **Volatility-weighted fees.** The schedule of equation (19) already rises into the open. Weighting it by realised session volatility rather than by elapsed time raises LP revenue in exactly the states where the short exposure is largest, and it is what moves the tolerance from 3.79% at $1\times$ turnover toward the 55.11% measured at $10\times$ turnover. It costs nothing when volatility is low, which is when the LP is not being paid for risk it is not taking.
4. **An external subsidy with an explicit sunset.** A protocol-funded subsidy is the only lever that can close the condition if flow turns out to be structurally one-sided, and the only one that can silently become permanent. It is therefore gated: it may be switched on only against a measured breach of the closure condition over a defined window, it is sized to the measured breach rather than to a target imbalance, and it carries a sunset date fixed at the moment it is switched on. A subsidy without a sunset is not a cold-start measure; it is a permanent transfer from the protocol to the liquidity side, and it converts a viability question into an accounting one.

The recommended sequence is 1–2–3 at launch, with 4 held in reserve and triggered only by measurement. If the measured imbalance stays inside the tolerance on the curated list, the product is viable without a subsidy and the question is settled by evidence rather than by argument. If it does not, the protocol learns the size of the gap in dollars before committing to close it.

6.5 The parameter set

Table 6. Parameter specification. Values marked *design* are the original specification; values marked *measured* are the recalibration of §7.

Parameter	Symbol	Value	Rationale and constraint
Leverage factor	λ_k	$1/Q_{1-\alpha}(M)$	Derived, not backtested. Per name and per session type, recomputed every session on a trailing window of 252–504 sessions chosen by forward error (§7.9); weekend and holiday pooled. The rounding lattice must be stated (0.25% reproduces the published values).
Saturation probability	α	1%	The single risk-appetite dial. It is a <i>frequency</i> budget; §7.5 shows it does not equalise severity across sessions.
Saturation point	$c = 1/\lambda$	implied	The maximum hedgeable gap and the maximum recovery as a fraction of notional. Beyond it the holder bears unhedged residual exposure.
Protocol fee	η_k	12% annualised	Prorated by term. 2.40 bp overnight, 8.97 bp weekend. Guardrail $\eta \leq 0.05 p_L$.
Trading fee	$\phi(t)$	0.10% $\rightarrow$ 1.00%	Linear rise into the open, compensating LP adverse selection, and scaled by realised volatility relative to the calibration reference (equation (20)).
Collateral	U	USDG (6 dec)	The chain's native stable unit, already integrated with lending venues. Decimals must be asserted at construction (§9.4).
Settlement source	—	official prints	Never a chain price for GAP. This is the property that makes settlement manipulation-proof.
Fallback rule	—	defer, not void	Revised. §8 shows the intuitive half-value void is a free butterfly; deferred settlement removes it at 0.021 bp.
Event session	C	separate	Own leverage and premium, calibrated on announcement sessions alone (§7.10). 1.0–1.4% of sessions, σ 3.3–4.7 $\times$ the non-event overnight, and leaving them in the overnight sample understates the premium by 4.0–16.5%. Not excludable from the product: an announcement is the event a hedger most wants covered.
Eligible names	—	canonical list	Only officially listed contracts. Names with a pending corporate action suspended for the affected session.

7. Empirical calibration

The design's method is sound; its inputs were invented. This section measures the quantity the protocol actually settles on.

7.1 Data and method

53,112 close-to-open gaps were computed for 22 US large caps over 2014–2026 from daily OHLC data obtained via a public historical quote API. The gap is $G = O/C - 1$. Sessions are classified by the number of calendar days the gap spans, which is exactly the design's own taxonomy: span 1 $\rightarrow$ E, span 3 $\rightarrow$ W, otherwise H. Truncated moments are computed on the raw gaps, so the cap is a true truncation rather than a mean of pre-truncated values. Raw pulls are cached so the analysis re-runs without re-fetching.

7.2 Measured session volatilities

Table 7. Session volatility: illustrative assumptions versus measured values. "Pooled" is the cross-sectional average; "high-beta" is the subset the product targets. Ratio is measured high-beta over design.

Session	Design	Measured, pooled	Measured, high-beta	Ratio
E overnight	1.60%	1.74%	2.18%	1.36×
W weekend	2.60%	1.78%	2.25%	0.87×
H holiday	2.00%	1.44%	1.82%	0.91×

THE THREE MATERIAL CORRECTIONS

Overnight volatility is understated by 36% for a high-beta name (2.18% measured against 1.60% assumed). The design's own target is a high-beta large cap, so this is the like-for-like comparison, and the error is in the unsafe direction.

Weekend volatility is overstated by 13% (2.25% against 2.60%). Measured, a weekend gap is the same size as a weekday overnight gap — not 62% larger as assumed.

The weekend variance ratio is 0.356, not 0.88. Per calendar day the weekend carries about a third of the weekday variance rate.

The sign of $\sigma_W - \sigma_E$ is not stable across names: 14 of 22 are weekend-fatter, 8 are overnight-fatter, and the split follows a mechanism rather than noise. The overnight-fatter group is mega-cap technology that reports earnings after the weekday close. This matters for the event-session term of §6.2: the "overnight is quieter" intuition holds for the index and fails for exactly the names a new product would launch on.

7.3 Consequences for leverage

Applying the design's own rule (14) to the measured sample, and then checking the property the rule is supposed to deliver — $P(\lambda|G| \geq 1) = \alpha$:

Table 8. Leverage recalibration at $\alpha = 1\%$. Realised saturation is measured under the design's leverage against the stated α .

Session	Design λ	Measured λ^* at $\alpha=1\%$	Realised P(saturate)	vs stated α
Overnight	19	12.1	2.735%	2.74×
Weekend	11	11.6	0.873%	0.87×

- **The rule is correct.** Fed measured inputs it delivers what it promises: $P(\text{saturate}) = 1.00\%$ overnight and 1.02% weekend against $\alpha = 1\%$.
- **Fed the design's inputs it does not.** At $\lambda = 19$ the overnight session saturates on 2.74% of sessions — $2.74\times$ the advertised rate. Over-levered overnight is a *safety* error.
- **The ordering is inverted.** The design gives the overnight session more leverage than the weekend (19 vs 11). Measured, the two are near-identical and the overnight session is the marginally riskier of the two (12.1 vs 11.6 at the same α). The design ranked them by σ ; the tail ranks them the other way.
- **Under-levered weekend is a competitiveness error** — capital efficiency left on the table, not a safety failure.

7.4 Pricing error: the distribution is misspecified

Equation (12) was evaluated three ways on the measured sample: the normal closed form, a variance-matched Student-t with $\nu = 4$, and the empirical truncated mean.

Table 9. Fair premium p_L under three distributional assumptions.

Session	p_L measured	p_L normal	p_L t($\nu=4$)	Normal error	t(4) error
Overnight	0.2173	0.3278	0.2815	+50.8%	+29.6%
Weekend	0.1445	0.1974	0.1733	+36.6%	+19.9%

Measured excess kurtosis is 30.7 on the overnight gap and 30.9 on the weekend, against zero for a normal by construction. Against the empirical distribution the normal overstates p_L by +51% overnight and +37% on the weekend. A Student-t with $\nu = 4$ is still not fat enough: it leaves a +30% and +20% residual. The design proposes to "fit both Gaussian and fat-tailed models"; the measurement says neither is adequate, and the fat-tailed candidate it named is not fat enough either.

This is a *pricing* error, not an insolvency one — Theorem 1 is unaffected. But p_L is the seed ratio, so a pool seeded at the design's value opens rich and the difference is a transfer to whoever arbitrages it first. The design's own algebra reproduces exactly ($p_L = 0.2282$ against a stated 0.228): the algebra is right, the distribution is wrong.

A CAVEAT THAT CUTS THE OTHER WAY

The empirical distribution is the correct reference for a *pricing* question and the wrong one for a *risk* question: an empirical sample cannot contain a gap larger than the worst one observed, so it understates tail risk by construction. The direction of this finding is safe; its magnitude is a lower bound.

7.5 Frequency versus severity

A single α is a frequency budget, not a severity budget. At a constant α , the residual the writer keeps beyond the cap, $E[(|G| - c)^+]$, differs materially by session:

Table 10. Residual exposure at a constant $\alpha = 1\%$.

Session	Cap c	P(saturate)	$E[(G - c)^+]$	Conditional severity
Overnight	8.25%	1.003%	0.0511%	5.10%
Weekend	8.65%	1.018%	0.0421%	4.13%
Holiday	6.26%	1.023%	0.0203%	1.99%

The rule delivers $\alpha = 1\%$ per session as intended, but the residual spread is **2.5×** across sessions: at the same α , the overnight session leaves the writer with 0.0511% of notional of expected unhedged exposure against the holiday session's 0.0203%. A constant α equalises how *often* the cap is breached, not how *much* is lost when it is. To equalise severity the overnight α would have to fall from 1.0% to 0.797% (cap 9.25%, λ 10.8). The cleaner fix is to state the risk appetite in the residual rather than in the frequency and derive α per session from it, making the choice a single number with a meaning rather than a frequency that silently carries different severity in different sessions.

7.6 On-chain depth and token mechanics

Whether the design is deployable depends on facts about the live chain, and these were read directly rather than assumed.

Table 11. Measured reference-market depth for three canonical tokens, all fee tiers combined, read through the Uniswap V3 factory at the fork block.

Pair	Fee tier	Pool	Depth
NVDA / USDG	500	0xd4EB...14a3	\$8,470,217
NVDA / USDG	3000	0xB944...7E2B	\$56,794
TSLA / USDG	3000	0xf4AC...89E3	\$1,330,357
TSLA / USDG	500	0xc4f0...0236	\$57,227
AAPL / USDG	500	0xAae0...2d6D	\$610,804
AAPL / USDG	3000	0x783C...b7Ed	\$143,595

Depth exists but is thin and uneven: **\$8.53M for NVDA, \$1.39M for TSLA, \$0.75M for AAPL**, and each is concentrated in a single pool. All three tokens have pools, so the named test universe is listable in principle. Two distinctions matter. BELL's own depth gate (10) is about BELL's *seeded* pool, not the reference market, and asks for $n \geq 3.52M$ tokens — a few hundred thousand dollars per market, which is achievable. Reference-market depth matters instead for the CAL extension (§10.2), whose manipulation-resistance argument requires an attacker to move every venue; at \$0.75M for AAPL that is a much lower bar than at \$8.5M for NVDA.

Four further facts about the live system, each of which changes a design assumption:

- **The venue set is real and enumerable.** Uniswap v2, v3 and v4 are all deployed and active on the chain — 25,243 pools and \$1.39B of 24-hour volume at the time of measurement, with the V3 factory at 0x1f7d7550B1b028f7571E69A784071F0205FD2EfA (49,073 bytes of code, returning real pools with real balances). This is not the canonical Ethereum address for any Uniswap deployment, a point that matters methodologically (§9.4).
- **The marking is to the reference, not to an independent 24/7 market.** Each token has its own price feed read through the standard aggregator interface, returning share price $\times$ multiplier, and the feeds "update 24/5, following market hours" — not 24/7. On-chain DEX trading is arbitrated *against* this feed rather than constituting a separate source of truth. The session gap is therefore a real P&L event for token holders, not an accounting artefact, and the demand for the hedge is genuine.
- **Corporate-action adjustment is live and dated.** Reading the multiplier interface at the fork block: TSLA at exactly 1.0 (never adjusted); NVDA at 1.000775159164630595 (+775.16 ppm, effective 2026-09-10); AAPL at 1.000566080061092436 (+566.08 ppm, effective 2026-08-14). Two of three sampled tokens are off parity, and both dates are consistent with quarterly dividend ex-dates. The issuer documents that the price oracle is paused while a corporate action is processed. This connects two previously separate findings: the corporate action is the mechanism that drives settlement into the branch §8 identifies as mispriced, so the risky branch fires on a schedule — roughly quarterly per name.
- **Minting and burning are unavailable for most of the weekend session.** The issuer supports mint and burn from Monday 02:00 CET to Saturday 02:00 CET; outside that window they are not supported. The weekend session the protocol prices runs Friday 16:00 ET to Monday 09:30 ET, which is almost entirely inside the no-mint window. On-chain trading is still permitted, so a hedger can trade, but any part of the de-

sign that assumes primary-market creation over the weekend assumes something the issuer does not support.

7.7 Measured leverage and premium table

Combining §7.2–7.5, the recalibrated parameter set for a high-beta name is:

Table 12. Recalibrated parameters. Design values are retained for comparison.

Parameter	Design	Measured	Direction of error
σ_E (high-beta)	1.60%	2.18%	Understated 36% — unsafe
σ_W (high-beta)	2.60%	2.25%	Overstated 13% — uncompetitive
σ_H (high-beta)	2.00%	1.82%	Overstated 9%
WVR	0.880	0.356	Weekend risk overstated
λ_E at $\alpha=1\%$	19	12.1	Over-levered — unsafe
λ_W at $\alpha=1\%$	11	11.6	Slightly under-levered
p_L overnight, held at $\lambda=19$	not published	0.2173	Gaussian form gives 0.3278, +50.8%
p_L weekend, held at $\lambda=11$	0.228	0.1445	Gaussian form gives 0.1974, +36.6%
σ_{ann} (design inputs)	29.61%	—	Naive scaling gives 25.4%

Two features of this table are worth stating plainly. The first is that the design's errors are not uniform: the overnight assumption is unsafe while the weekend assumption is merely uncompetitive, and because the design ranked leverage by σ the two errors *compound* into an inverted ranking. The second is that the Gaussian pricing error is an order of magnitude larger than the fat-tail correction the design anticipated, which means the pricing framework — not just its inputs — needs revision before launch. §7.11 makes that revision and prices it: a fitted normal-inverse-Gaussian removes almost all of the error, at a cost that keeps the fit off-chain. §7.9 answers the other half of the question, the window the quantile is estimated over, and §7.10 the session the taxonomy omits.

7.8 Per-name calibration

Everything above is pooled or class-level. §5.3 requires a per-name, per-session calibration, and a pooled number cannot supply it: the cross-section is wide enough that a class-level σ is a starting point rather than a calibration. The table below is the per-name set, with standard errors produced by resampling the gaps and *re-running the leverage rule inside each resample*. That detail matters. λ^* is an order statistic, so holding it fixed and reporting a standard error on p_L would understate the uncertainty on every premium in the table — which is the class of error this section exists to stop making.

Table 13. Per-name, per-session calibration for the three canonical names. λ^* is the lattice leverage, $c = 1/\lambda^*$ the cap, $p_L = \lambda^* \cdot E[\min(|G|, c)]$ the fair premium. SE is the bootstrap standard error of p_L over 2,000 resamples with the leverage rule re-estimated inside each; CV is SE/p_L .

Name	Session	n	σ	$Q_{0.99}(G)$	λ^*	c	p_L	SE(p_L)	CV
NVDA	E overnight	1,965	1.88%	6.34%	15	6.67%	0.1740	0.0125	7.2%
NVDA	W weekend	452	2.15%	8.96%	11	9.09%	0.1413	0.0286	20.2%
NVDA	H holiday	96	1.62%	4.37%	22	4.55%	0.2555	0.0537	21.0%
TSLA	E overnight	1,965	2.18%	8.76%	11	9.09%	0.1465	0.0075	5.1%
TSLA	W weekend	452	2.57%	9.69%	10	10.00%	0.1693	0.0323	19.1%
TSLA	H holiday	96	2.72%	8.37%	11	9.09%	0.1799	0.0464	25.8%
AAPL	E overnight	1,965	1.09%	4.35%	22	4.55%	0.1440	0.0108	7.5%
AAPL	W weekend	452	1.45%	6.10%	16	6.25%	0.1306	0.0308	23.6%
AAPL	H holiday	96	1.12%	3.09%	32	3.13%	0.2356	0.0516	21.9%

The bootstrap separates the sessions cleanly, and the separation changes the recommendation. Overnight is well determined: CV(p_L) of 5.1–7.5% on roughly 1,965 observations, with the tightest cell — TSLA overnight — giving a 95% interval of [0.1325, 0.1617]. Weekend and holiday are not: CV of 19.1–25.8%, with SE(λ) as large as 8.1 on a point estimate of 32. Publishing a per-name weekend λ would be publishing noise. AAPL's holiday λ of 32 ± 8 is barely distinguishable from TSLA's 11 ± 3.6 , and both rest on 96 observations. The per-name calibration should therefore be applied *to the overnight session only*; weekend and holiday should be pooled across names, or shrunk toward a class mean, until the sample supports more. This is the same conclusion the forward-error test of §7.9 reaches by an entirely different route, and the estimability test below reaches it a third time, which is some evidence that it is a property of the data rather than of the method.

One further number belongs here because it constrains what a per-name launch list can promise. The per-name λ spans 10 (TSLA weekend) to 32 (AAPL holiday) — a factor of 3.2 on nominally the same product. A single class-level λ is not merely imprecise; it is wrong by more than a factor of three at the extremes, and it is wrong in a direction that depends on the name.

The bootstrap settles the recommendation. It does not yet state a criterion, and without one the same question returns for every session type and is answered case by case — which is how a calibration drifts. The criterion is short. The leverage is $\lambda^* = 1/Q$, and Q is an order statistic, so its error is governed not by the sample size but by the number of observations at or beyond the quantile being estimated: $n(1 - \alpha)$, which at $\alpha = 1\%$ is one observation in a hundred. When a sample holds no draw in the top per cent, the empirical “99th percentile” is really the sample maximum, which is a lower quantile. The estimator is therefore biased *low* on Q and so biased *high* on λ . The lattice then fixes the tolerance exactly, because λ is floored to integers: adjacent published values differ by one, so a standard error measured in λ is already denominated in lattice steps, and an estimate whose error exceeds a step cannot place its own value.

Table 14 applies that test to all four session types. The operative column is the third: at $\alpha = 1\%$ the overnight session has 19.7 expected tail observations, the weekend 4.5, the holiday 0.96 and the event session 0.34. Everything else in the table follows from that column, and so does the recommendation.

Table 14. Whether a per-name leverage can be published, by session type. The governing quantity is the expected number of observations at or beyond the 99th percentile, $n(1 - \alpha)$, and not the sample size. $SE(\lambda)$ is quoted in lattice steps, which are exactly 1 in λ . Signal-to-noise is the spread of λ across the three canonical names divided by the mean standard error. Standard errors are the bootstrap values of Table 13; the event session is a Gaussian-reference figure, which its measured near-Gaussian kurtosis makes the right reference there (§7.10).

Session	n	E[tail obs]	P(none beyond $Q_{0.99}$)	$SE(\lambda)$, lattice steps	Signal-to-noise	Verdict
E overnight	1,965	19.65	0.0%	0.67–2.00	4.3	per-name; publish the interval
W weekend	452	4.52	1.1%	2.24–4.69	1.0	pool
H holiday	96	0.96	38.1%	3.55–8.08	1.9	pool
C event	34	0.34	71.1%	0.82–1.80	3.0	pool — biased, not noisy (§7.10)

Three consequences sharpen the recommendation rather than repeat it. First, the overnight session is the only one whose error is of the order of a single lattice step, and it is publishable — but the interval has to be published with it. AAPL's overnight λ of 22 carries a standard error of 2.00, which spans four lattice points; a point estimate quoted without that width overstates what 1,965 observations buy. Second, the weekend fails on a different ground than its bootstrap CV suggests. Its error is 2.2 to 4.7 steps, but the more damaging number is the signal-to-noise ratio of 1.0: the spread of λ across the three canonical names is no larger than the error attached to each of them. A per-name weekend λ is not a noisy estimate of a real quantity; on this sample it is an estimate of nothing, and pooling is not a compromise but the only defensible reading. Third, the event session fails for a reason the other two do not share, and the distinction matters. Its spread, at 0.82 to 1.80 steps, is of the same order as the overnight session's and better than the weekend's, so it is not noise that disqualifies the empirical route there. It is a one-signed bias of 13.8 per cent on the quantile, which does not average out across a launch list. §7.10 specifies the calibration that removes it.

7.9 The leverage window, chosen by forward error

§6.1 sets leverage from a quantile of the absolute gap "with a model-based tail extension", and §5.3 says the extension is used "where a name's sample is too thin to place the truncation point at all". Both sentences name a mechanism and neither names a window, and the window is the parameter that decides whether the rule works. A trailing quantile is useful only if it forecasts the quantile that will actually be realised, so the window is selected by exactly that test: for each session in the sample, the leverage implied by the trailing w sessions is scored against the 99th percentile realised over the next w sessions, and the figure reported is the mean absolute relative error of that forecast.

Table 15. Forward error of the trailing 99th percentile, by window length. Each cell is the mean absolute relative error of a leverage derived from a trailing window of that many sessions, scored against the 99th percentile realised over the following window. A dash means too few non-overlapping forecast pairs to estimate. Bias is the mean ratio of the trailing quantile to the realised one; a ratio above one means the rule under-levers, which is the safe direction.

Name	Session	n	126	252	378	504	Best	Bias
NVDA	E overnight	1,965	42.1%	26.7%	23.6%	16.9%	504	0.98
TSLA	E overnight	1,965	36.7%	32.8%	24.4%	20.7%	504	1.10
AAPL	E overnight	1,965	45.1%	29.3%	28.8%	32.9%	378	1.09
NVDA	W weekend	452	53.9%	—	—	—	126	1.13
TSLA	W weekend	452	63.0%	—	—	—	126	1.21
AAPL	W weekend	452	122.6%	—	—	—	126	1.75

The window is answered by the table, and the answer is the opposite of the intuition. On the overnight session, longer is better and monotonically so for two of three names: NVDA falls from 42.1% at 126 sessions to 16.9% at 504, TSLA from 36.7% to 20.7%, and AAPL from 45.1% at 126 to 28.8% at 378, rising again to 32.9% at 504. A short window is not more responsive, it is more wrong, because the object being estimated is a 99th percentile and a 126-session window contains roughly one observation in the tail it is trying to locate.

The weekend session cannot support a per-name window at all. There are 452 weekend observations for the whole sample, so 126 sessions is the longest window with more than a handful of forecast pairs, and the error at that window is 53.9% for NVDA, 63.0% for TSLA and 122.6% for AAPL. This is the same conclusion §7.8 reached from the bootstrap standard errors, arrived at by an entirely different route, which is the main reason to believe it is a property of the data rather than of either method.

The bias direction is worth recording because it is a safety margin rather than a defect. At 126 sessions the trailing quantile is biased high on every overnight cell — ratios of 1.15, 1.07 and 1.14 — so the rule under-levers and the protocol's exposure is smaller than intended. The bias moves toward one as the window lengthens: 0.98, 1.10 and 1.09 at the best window for each name. The exception is AAPL's weekend cell at 1.75, which is the one place where the trailing quantile sits far enough above the realised one to be a competitiveness problem rather than a safety margin: the pool would be persistently under-levered against a premium it could charge.

The model-based tail extension is specified, tested, and not adopted as the default. Extending the fitted NIG tail below the sample minimum is compared with the raw empirical quantile over the same forward windows, and it helps one name while hurting two: NVDA overnight improves by 51.1%, while TSLA overnight degrades by 20.8% and AAPL overnight by 10.0%. A procedure that is right on one of three canonical names is not a default; it is a fallback, and it is adopted here as one — used only where the window is too short to place the quantile at all, with the fact recorded in the parameter set so the premium stays auditable.

One further check belongs here because §12.2 raises it. The design's stationarity warning is the regulatory record of a 22.5% year-over-year rise in trading pauses, which, if it appeared as a drift in the quantile, would argue for a shorter window. It does not appear as one: regressing the rolling 99th percentile on time gives slopes per hundred sessions of +1.07%, -0.56% and -1.32% on the overnight session and -2.42%, -12.67% and +21.19% on the weekend, with the signs disagreeing across names. There is no common drift to chase. The recommendation is therefore a trailing window of 252 to 504 sessions on the overnight session, re-estimated every session, with a minimum observation count below which the fitted extension is used and tagged; weekend and holiday pooled; and the window itself re-tested on a schedule rather than fixed permanently, because a warning that has not yet shown up in this sample is not the same as a warning that is wrong.

7.10 The event session, measured

§6.2's taxonomy is E, W and H: sessions distinguished by length. There is a fourth session that length does not capture and the design names without pricing. An announcement gap is a single overnight session carrying the firm's entire information release; it is E by duration and a different object by distribution. The measurement below uses 936 real earnings dates for 22 names, retrieved from the Nasdaq calendar and matched to the gap that follows each announcement.

Table 16. The event session, measured. C is the session containing the first print after an announcement. σ_C and σ_{nonC} are the conditional volatilities; λ and p_L are the leverage and fair premium computed on the whole sample and on the sample with C excluded, at $\alpha = 1\%$. Δp_L is the error introduced by calibrating on a sample that contains events.

Name	Events	Share	σ_C	σ_{nonC}	Ratio	λ all	λ excl C	p_L all	p_L excl C	Δp_L
NVDA	34	1.35%	7.93%	1.68%	4.73×	14	17	0.1654	0.1927	+16.5%
TSLA	35	1.39%	7.52%	2.11%	3.56×	11	12	0.1548	0.1609	+4.0%
AAPL	35	1.39%	3.64%	1.09%	3.34×	21	25	0.1439	0.1614	+12.1%
Universe, 22 names	16–35	0.96–1.39%	1.95–10.43%	0.61–3.06%	1.71–8.88×	—	—	—	—	—

Announcement sessions are rare and large. They are 1.0 to 1.4% of sessions across the universe, and the volatility in them is 3.34 to 4.73 times the non-event overnight volatility for the canonical three, with a universe range of 1.71 to 8.88 and a median of 4.16. The design's claim that the C session is unpriced is correct, and the direction of the error is knowable: calibrating the overnight session on a sample that includes events raises the 99th percentile, which lowers λ^* , and the lower leverage more than offsets the larger moment. λ falls from 17 to 14 for NVDA, from 12 to 11 for TSLA and from 25 to 21 for AAPL — one to four lattice points — and the fair premium falls by 16.5%, 4.0% and 12.1% respectively. The pool therefore opens cheap, and the transfer runs to whoever buys the cheap leg.

The recommendation follows the mechanism rather than the magnitude. Excluding event sessions from the overnight calibration and treating C as its own session with its own leverage and premium removes a 4 to 16% pricing error that is otherwise invisible. C cannot simply be pooled: at 1.4% of sessions it contributes 34 or 35 observations per name, too few to calibrate alone and too influential to ignore. It also cannot be excluded from the *product*, because an announcement is exactly the event a hedger most wants covered — so it is a separate parameter set, not a missing one.

Two details of the measurement are stated so the number can be audited. The reporting convention is inferred per name rather than per event, from the ratio of mean absolute gap out of versus into the announcement date, which resolves all three canonical names to after-the-close with ratios of 7.2, 6.3 and 4.6; inferring it per event would select on the outcome being predicted. And the calendar was retrieved for 2014-01-01 to 2025-05-05; 355 later dates returned HTTP 403 and could not be recovered, so the event share is measured on 29 of 46 announcements per name and is a lower bound. The direction of the finding does not depend on the missing tail, but the share does.

Table 14 rules out a per-name empirical quantile for C. It does not rule out a per-name calibration, and the distinction is the whole of the specification. A per-name quantile conflates two quantities: the scale of the session and the shape of its distribution. The measurements say the two have opposite characters, so they should be estimated by different means.

The shape is homogeneous, and near-Gaussian. Measured excess kurtosis in the event session is -0.08 ± 0.48 across the canonical names (NVDA +1.92, TSLA -0.96 , AAPL -1.19), against 13.24 ± 0.06 for the non-event pool, whose per-name values run from 9.11 to 20.87. Only one of the three names has an event-session kurtosis distinguishable from zero at 5 per cent, and that is the weakest form of the evidence. The contrast is the finding: *the event session is high-variance but not fat-tailed*. Conditioning on a scheduled announcement removes the surprise, so the session is drawn from a wide but well-behaved distribution, while the fat tail lives in the non-event sessions — the unscheduled macro gaps that arrive without warning. This inverts the natural expectation, and it has a direct architectural consequence: the fat-tail machinery of §7.11 is required for the

non-event pool and is not required for C. A parametric quantile is unusually well justified precisely where the design was least able to estimate one empirically.

The scale is name-specific, and it is estimable. The event multiplier $r = \sigma_C / \sigma_{\text{non-C}}$ can be estimated at 34 observations even though a quantile cannot, because the name's own level is pinned by roughly 2,479 non-event observations and only the ratio carries event information. Across the 22-name universe that ratio spans 1.71 (XOM) to 8.88 (NFLX), and its cross-sectional spread, $\tau = 1.58$, exceeds the within-name sampling error by a factor of 6.7. Shrinkage toward the pooled mean is therefore mild — weights of 0.67 to 0.98, with only the extremes moving materially (NFLX $8.88 \rightarrow 7.38$, XOM $1.71 \rightarrow 1.76$). The honest description is not that shrinkage rescues a weak estimate, but that the estimate was never the weak part.

The specification is therefore: *pool the shape, keep the scale*. λ_C is placed from the pooled shape applied to the name's own shrunk event scale, and not from a per-name empirical quantile on 34 observations. The shape constant is measured, not assumed: across the universe the ratio of the 99th percentile of $|G|$ to σ in the event session has a median of 2.294, which is 11.0 per cent *below* the Gaussian 2.576 — the same thin-tailed character the kurtosis reports, seen from the quantile side. Applied to the shrunk scales it puts λ_C at 5 for NVDA, 5 for TSLA and 11 for AAPL, and across the 22-name universe it ranges from 4 to 21 with a median of 6. The reason to prefer it is not tidiness but bias. A per-name empirical quantile at $n = 34$ is biased low on Q by 13.8 per cent with a spread of 16.4 per cent. Those are measured, by drawing 34 observations from the pooled standardised event sample — 719 observations, 7.2 of them expected in the tail — and scoring the estimator against the population quantile, so the population is held fixed and only the sample size changes. The bias is one-signed and therefore does not average out across a launch list. Its size in lattice steps depends on the name, because the step is exactly 1 in λ and λ_C itself ranges from 4 to 21: it is +0.69 steps for NVDA and TSLA, +1.52 for AAPL, and up to +2.89 for the lowest-volatility names, where λ_C is largest. The parametric route removes it. The same test returns a bias of -0.4 per cent and a spread of 12.6 per cent, which is the error on σ_C itself. This is a narrower claim than it first appears, and it is the answer to the objection that 34 observations are too few: the sample cannot support a per-name event *quantile*, but it comfortably supports a per-name event *scale*.

The calibration is only useful if it is published name by name, and the objection that 34 observations cannot support one is answered by the shape and the scale separating cleanly. Table 17 is the resulting parameter set: the multiplier measured and shrunk, the event scale it implies, the leverage that follows, and the premium at that leverage.

Table 17. The event-session parameter set, by name. r is the measured multiplier $\sigma_C/\sigma_{\text{non-C}}$ and r^* its shrunk value; σ_C^* is the implied event scale; $\lambda_C = \text{floor}(1 / (q_C \cdot r^* \cdot \sigma_{\text{non-C}}))$ with the interval coming from the uncertainty on r^* alone; and $p_{L,C} = \lambda_C \cdot E[|G|_C \leq c]$ is the event premium at that leverage, with its bootstrap standard error. The pooled shape constant is $q_C = 2.294$.

Name	n_C	r	SE(r)	r^*	σ_C^*	λ_C	interval	$p_{L,C}$	SE($p_{L,C}$)	CV
AAPL	35	3.341	0.399	3.402	3.70%	11	10–13	0.3613	0.0569	15.7%
AMD	34	3.938	0.478	3.974	7.96%	5	4–6	0.3171	0.0567	17.9%
AMZN	35	5.424	0.648	5.275	6.33%	6	6–7	0.3360	0.0599	17.8%
AVGO	34	3.452	0.419	3.511	5.13%	8	7–9	0.3071	0.0536	17.5%
COIN	16	2.856	0.505	2.994	9.16%	4	4–5	0.2270	0.0760	33.5%
CRM	34	5.715	0.693	5.502	6.76%	6	5–7	0.3366	0.0579	17.2%
GOOGL	35	5.308	0.634	5.180	5.46%	7	7–9	0.3339	0.0598	17.9%
INTC	35	4.522	0.540	4.507	7.41%	5	5–6	0.3179	0.0571	18.0%
JPM	35	1.996	0.239	2.048	2.27%	19	17–21	0.3480	0.0573	16.5%
KO	35	3.033	0.363	3.099	2.17%	20	18–22	0.3455	0.0568	16.4%
META	35	8.089	0.967	7.091	9.15%	4	4–5	0.3148	0.0629	20.0%
MSFT	35	3.868	0.462	3.907	3.88%	11	10–12	0.3734	0.0591	15.8%
MU	35	3.531	0.422	3.586	7.00%	6	5–7	0.3423	0.0576	16.8%
NFLX	35	8.883	1.062	7.499	8.58%	5	4–5	0.3704	0.0643	17.3%
NVDA	34	4.729	0.574	4.689	7.86%	5	4–6	0.2821	0.0590	20.9%
ORCL	34	5.432	0.659	5.278	7.28%	5	5–6	0.3121	0.0585	18.8%
PG	35	4.487	0.536	4.475	2.72%	16	14–18	0.3796	0.0566	14.9%
PLTR	19	4.432	0.719	4.422	10.14%	4	3–5	0.2957	0.0835	28.3%
QCOM	35	4.375	0.523	4.375	6.06%	7	6–8	0.3523	0.0575	16.3%
SMCI	24	3.493	0.504	3.573	9.19%	4	4–5	0.3160	0.0721	22.8%
TSLA	35	3.557	0.425	3.611	7.63%	5	5–6	0.3192	0.0584	18.3%
XOM	35	1.714	0.205	1.757	2.00%	21	19–24	0.3363	0.0552	16.4%

Three things are worth reading off it. First, λ_C spans 4 to 21, a wider range than the overnight leverage's 11 to 22, and it is *low* almost everywhere: a median of 6 against the overnight median of 15. The event cap therefore sits near 20 per cent on most names and the claim is close to linear in the gap, so the truncation that does real work in the fat-tailed non-event sessions does very little here. Second, that is what makes the event premium expensive. $p_{L,C}$ runs 0.227 to 0.380 with a median of 0.335, against 0.1440 to 0.1740 for the overnight session on the canonical names: covering an announcement costs roughly twice what covering an ordinary overnight costs. That is a fair price for the risk, and it is a number the launch plan should carry rather than discover. Third, these premiums are less well determined than the overnight premiums, with CVs of 14.9 to 33.5 per cent against 5.1 to 7.5. That is a property of the session and not of the method — the event sample is 34 observations wherever the overnight sample is 1,965.

Is the shrinkage stable out of sample? τ is estimated from 22 names, so it is itself an estimate, and the question is fair. Three tests, and they agree. A jackknife over the universe puts τ between 1.290 and 1.643 with a standard deviation of 0.086, or 5.4 per cent of the estimate, so no single name drives it. Splitting each name's events into two disjoint halves gives a correlation of 0.850 between the halves' multipliers, which by Spearman-Brown implies a full-length reliability of 0.919: a name's own ratio measured on half its events predicts the other half, so the per-name signal is real rather than one draw dressed up as a parameter. And fitting

the pooled mean on half the names to predict the other half, scored against each held-out name's own independent second half, the shrunk estimate beats the raw one by 8.0 per cent.

The last test answers the concern directly. τ is noisy in the sense that any estimate from 22 observations is noisy, but the weight it produces is not, because the weight is $\tau^2/(\tau^2 + \text{SE}^2)$ and τ^2 exceeds SE^2 by a factor of about seven. Halving τ to 0.80 still leaves a mean weight of 0.655; doubling it to 3.19 raises the weight only to 0.968. Across a fourfold range of τ^2 the weight stays inside 0.66 to 0.97, so the specification is not sensitive to the parameter the objection is about. What would change the answer is a different standard error — a name with half the event observations — and that is a data-collection question rather than an estimation one.

7.11 The fitted alternatives, and what they cost

Section 5.3 said the seed model should be an empirical truncated distribution with NIG or jump-diffusion as fitted alternatives, and left both unfitted. Both are fitted here by maximum likelihood to each name and session, and priced at the *same* λ so that the comparison isolates the distributional assumption and nothing else.

Table 18. Fair premium p_L under each distributional assumption, at the per-name leverage of Table 13. The parenthesised figure is the error against the empirical truncated mean. Gaussian and $t(4)$ are shown for reference; NIG and Merton jump-diffusion are the fitted alternatives.

Name	Session	Empirical	Gaussian	$t(4)$	NIG	Merton-JD
NVDA	E	0.1740	0.2247 (+29.1%)	0.1964 (+12.9%)	0.1777 (+2.1%)	0.1812 (+4.1%)
NVDA	W	0.1413	0.1883 (+33.2%)	0.1654 (+17.1%)	0.1439 (+1.9%)	0.1456 (+3.0%)
NVDA	H	0.2555	0.2837 (+11.1%)	0.2455 (−3.9%)	0.2553 (−0.1%)	0.2557 (+0.1%)
TSLA	E	0.1465	0.1914 (+30.6%)	0.1681 (+14.7%)	0.1464 (−0.1%)	0.1492 (+1.8%)
TSLA	W	0.1693	0.2054 (+21.3%)	0.1801 (+6.4%)	0.1717 (+1.4%)	0.1733 (+2.4%)
TSLA	H	0.1799	0.2386 (+32.6%)	0.2081 (+15.7%)	0.1779 (−1.1%)	0.1849 (+2.8%)
AAPL	E	0.1440	0.1913 (+32.8%)	0.1680 (+16.7%)	0.1453 (+0.9%)	0.1492 (+3.6%)
AAPL	W	0.1306	0.1857 (+42.1%)	0.1632 (+24.9%)	0.1337 (+2.4%)	0.1400 (+7.2%)
AAPL	H	0.2356	0.2846 (+20.8%)	0.2462 (+4.5%)	0.2336 (−0.8%)	0.2392 (+1.6%)

The result is the one the framework needs. A Gaussian seed prices the overnight session 29 to 42 per cent rich, which is the arbitrage transfer §7.4 identified. **NIG prices it within −1.1 to +2.4 per cent.** Merton jump-diffusion lands at +0.1 to +7.2 per cent, and $t(4)$ — the family the design named as its fat-tailed reference — still overprices by 4.5 to 24.9 per cent, which is why §5.3 rejected it. The misspecification is not merely diagnosed; it is closable, and the family that closes it is one the design named but never fitted. By likelihood NIG is also the best of the four — a log-likelihood of 5,419.7 on 1,965 NVDA overnight observations against 5,400.9 for Merton-JD, 5,376.2 for $t(4)$ and 5,023.4 for the Gaussian — so the pricing ranking and the fit ranking agree.

Two caveats belong in the main text rather than a footnote. First, maximum likelihood fits the likelihood, not the variance: the fitted NIG reproduces the sample standard deviation only to within 2–6 per cent and slightly understates it, which is why Merton-JD occasionally prices above NIG despite fitting worse. Second, a parametric tail is a model, and Table 18 is the evidence that a model can be much better than the Gaussian without being right.

What the fat tail costs on-chain. The architectural question is not how to fit NIG but whether the protocol can evaluate it. The terminal payoff is computed on-chain today because the normal truncated moment has a closed form in ϕ and Φ , and those are cheap enough to evaluate in the contract. NIG has no such closed form: its trun-

cated moment is an integral, so an on-chain implementation would have to integrate numerically. Measured on the deployment toolchain, the closed-form normal moment costs 37,439 gas; an NIG quadrature costs 21,400 gas *per term* and scales linearly, reaching 685,590 gas at 32 terms and 5.48 million at 256 — 18× and 146× the closed form respectively. Reading a published premium from storage costs 2,640 gas, roughly 260× less than the cheapest credible quadrature.

The conclusion is architectural and it is not close. **The fat-tailed fit belongs off-chain.** The oracle should publish the fitted premium as a committed parameter and the on-chain contract should not attempt to reproduce the fit. This changes what the oracle is trusted for — from publishing a volatility that the contract transforms, to publishing the transformed result directly — and that change should be stated in the design rather than left implicit in the implementation.

Stating it is not the same as specifying it, and the specification is short because the mechanism already exists in this paper. The trust shift is policed by route R5 of the settlement table — the optimistic challenge window, costed at 1.37 bp per session, monotone and free-option-free — which is priced in the settlement analysis and currently wired to nothing. Connecting it to the premium publisher adds no new cost, because the 1.37 bp is already carried. Four parts:

- **Commitment before the session.** The publisher posts the parameter set for session t no later than the close of session $t - 1$, as a hash over both the parameters and the raw inputs the fit consumed. It cannot fit after seeing the outcome, and because the inputs are committed too a challenge is verifiable rather than a matter of testimony.
- **A staleness bound.** A parameter set is usable only within a stated number of sessions of its commitment. Beyond the bound the pool refuses to price rather than pricing on a stale fit — the rule the BELL-IV provenance stamp already carries, applied to the premium rather than to the volatility.
- **A bonded dispute.** Any party may challenge within the window by posting a bond together with the committed inputs. The challenge resolves by re-running the published fit, which is deterministic given those inputs, so the dispute has a computable answer and does not require anyone to be believed. A frivolous challenge forfeits the bond to the publisher; a correct one forfeits the publisher's bond.
- **A fallback, not a halt.** While a challenge is open the pool prices on the trailing-realised estimate — the named fallback already specified for a stale volatility — so a dispute degrades the price rather than stopping the market. This is what makes R5 monotone with no free option in the settlement table, and it is the property that makes it safe to wire here.

This belongs in the design rather than in an operations runbook because the off-chain fit is the largest new trust assumption the pricing revision introduces, and an assumption without a policing mechanism is only a promise. With R5 wired, the publisher's discretion is bounded by a bond and a deterministic re-run, and the worst case is a session priced on the fallback rather than a session priced wrong.

Stating it is not the same as specifying it, and until now the specification was prose: four parts were named and not one carried a number. Each is derived here from a quantity this paper has already measured, and Table 19 collects them.

Table 19. The route R5 parameter set. Each value is produced by `r5_params.py` from data this paper has already measured, so none of them is a choice. The derivations are given in the text below.

Parameter	Value	Derivation
Staleness bound	12 sessions	The horizon at which the expected mispricing from a stale parameter reaches the half-spread, 1 per cent of notional. Measured per session type; the holiday session binds at 12, the overnight at 29 and the weekend at 40.
Publisher bond	\$500,000	3× the largest one-session mispricing gain, \$94,849 on NVDA. The raw figure is \$284,548 and it rounds up, because a publisher bond is a lower bound.
Challenger bond	\$50,000	$B_{\text{pub}} \cdot p/(1 - p)$ at a 10 per cent prior that the publisher is wrong, which is \$55,556 and a ratio of 10×. It is an upper bound on what a guessing challenger should risk, so it rounds down.
Publisher identity	1 key per name	The bond is per name, so one compromised key costs one name's bond and not the book's. A single key publishing the whole universe would make the bond the wrong size by the number of listed names.
Rotation period	12 sessions	The staleness bound, because beyond it a fresh commitment is required anyway and the refresh point is therefore the natural rotation point.
Bond lock after rotation	13 sessions	The rotation period plus the challenge window, so an outgoing publisher stays slashable for the whole time its last commitment can still be priced on.
Fallback window	504, 378 for AAPL	The overnight calibration window that §7.9 already selected by forward error. The fallback is the same estimator with the publisher removed, so it inherits that window rather than needing one of its own. The weekend window is 126, the longest its sample supports.

The staleness bound is not a median drift, and that is the first finding. Two estimators are candidates for the published leverage and they go stale in opposite ways. The empirical quantile over a fixed window is a *jump* process rather than a drifting one: the 99th percentile of a 252-session window is placed by roughly its top three points, so a window that slides by a single session usually retains exactly the same top three and returns exactly the same λ . Measured on NVDA overnight, the median one-session move is **exactly 0.000 steps at every horizon out to forty**. A bound set on that statistic would have been set on a number that is identically zero, which is why it is set instead on the shape-standardised estimator $\lambda = 1/(z\sigma)$. That estimator drifts smoothly, because σ does, and §7.10 established that z is the homogeneous part across names and sessions while σ is the heterogeneous one. Carrying the drift through σ is therefore the right reduction and the conservative one: it moves sooner in aggregate than the jump process does.

The jump process is still real, and it is frequent rather than slow. The published integer for NVDA overnight changes with probability **2.6 per cent in a single session**, 34 per cent within twenty and 88 per cent within a hundred and twenty. So the parameter does move; it moves in steps. That is the same lattice argument §7.8 makes about the *precision* of the estimate, arriving here as a statement about its *durability*, and it is why the bound is expressed in sessions of the same type: a weekend commitment is measured against weekend sessions, not calendar days.

The economic crossing governs rather than the lattice crossing. The lattice crossing says only that a refresh would begin to help; the half-spread is the error the protocol is willing to tolerate, and it is the same tolerance the maker's own quote is held to at 2 per cent of mid. On the full one-year window the overnight session binds at 29 sessions and the weekend at 40. The holiday session has 96 observations and cannot support that window at all, so it is measured on the widest one its sample allows, which makes the drift estimate noisier and biases the bound down. It binds at **12**, and 12 is adopted: a shorter refresh is the safe direction and it costs nothing, because the publisher commits per session regardless of the bound.

Table 20. The staleness measurement, by session type. The lattice crossing is the horizon at which the fit has moved one whole step in λ ; the economic crossing is the horizon at which the expected mispricing reaches the half-spread. The economic figure governs. The holiday row is measured on the widest window its 96 observations support rather than the one-year window, which makes its drift estimate noisier and biases the bound down rather than up.

Session	n	Window	Lattice	Economic	Binding
Overnight	1,965	252	22–47	29–42	29
Weekend	452	252	33	40–137	40
Holiday	96	60 (reduced)	13–23	12	12

The bond is sized by an identity rather than an assumption. The derivative of the premium in the leverage is exactly the truncated first moment, $dp/d\lambda = E[|G| \cdot 1\{|G| \leq 1/\lambda\}]$, which is what one lattice step of mispricing is worth per unit of notional. Evaluated at the published leverage and multiplied by the depth the pool commits to, it gives the most a publisher can take from a one-session mispricing: 1.1119 per cent of notional for NVDA at λ 15, on \$8.53M of committed depth, or \$94,849. Three times that is \$284,548, which rounds up to **\$500,000** and covers 5.3 such sessions.

The challenger bond rounds the other way, and the direction is the design. It is not a lower bound on what a challenger must risk; it is an upper bound on what a party who is only guessing should risk. Setting it at $B_{\text{pub}} \cdot p/(1 - p)$ leaves a challenger holding a 10 per cent prior with positive expected value, so a challenge remains worth mounting on a suspicion but not worth mounting on a coin flip. Anything larger would tax the verification this mechanism depends on, which would be a strange way to secure it.

The fallback window is not a new parameter, and the reason is worth stating. The fallback is not a second estimator needing its own window: it is the same trailing quantile the calibration uses, computed by a rule nobody controls. It therefore inherits the window §7.9 selected by forward error rather than needing a test of its own, and choosing a different window here would make it a different object. What the new work adds is the decomposition that explains *why* §7.9 came out the way it did. The fallback trades sampling noise against staleness: noise falls as $1/\sqrt{W}$, and the bias grows with the drift between the window's midpoint and the present. Both terms are measured rather than assumed, and they are the same order across the whole range. For NVDA the noise falls from 5.02 lattice steps at 126 sessions to 2.51 at 504 while the bias rises from 1.59 to 3.45, so the total is nearly flat at about six steps and no window is sharply better than its neighbours. That is the explanation for §7.9's result rather than a replacement for it, and it says something the forward-error table could not: the fallback is coarse, at roughly 4.7 times the committed fit's own sampling error of 1.27 steps, and it is coarse because the object being estimated is a 99th percentile rather than because the window was chosen badly. A degraded mode that is measurably degraded is a better thing to ship than one that is assumed to be adequate.

The mechanism is implemented rather than described. PremiumRegistry.sol holds the commitment, the staleness bound, the bond, the dispute and the fallback, and refuses any commitment for a session that has already opened — which is what enforces "no fit after the outcome" without a calendar. The arbiter's power is deliberately narrow and stated: it cannot alter a parameter, only rule on whether a committed one matches its committed inputs, which is a far weaker assumption than "the publisher will be honest" and is the one that remains. Sixteen tests cover the timing, the bound and its boundary, the dispute in both directions, the fallback, and the guards that stop a commitment without a bond or a parameter of zero from ever being priced on.

8. Settlement integrity

Theorem 1 covers the map from prints to payoff completely, and says nothing about the map from the world to the prints. That is the protocol's entire residual risk, and this section treats it as the central open problem.

8.1 The attacker's problem

Define an *intervention* as any action an adversary can take that changes the realised pair (C, O) . The relevant problem is

$$\text{maximise } \Pi_L(\text{after}) - \Pi_L(\text{before}) - \text{cost}(\text{intervention}) \quad \text{s.t. the intervention is available to a non-privileged party} \quad (23)$$

Three structural facts make this tractable, and each is a result rather than an assumption.

- **Fact 1 — the problem is one-dimensional.** Π_L is an even function of the gap, so direction carries no value; only the magnitude of the induced change matters. Verified through the real settlement function over $\pm 50\%$ at a residual of $0.0e+00$ for every session type (check S1).
- **Fact 2 — the prize is Lipschitz-bounded and the bound is tight.** $|\Delta \Pi_L| \leq \min(\lambda |\Delta A|, 1)$, measured at a ratio of 1.000000 over $60,000$ random pairs (check S3), so the bound is attained rather than merely valid. Two consequences follow for free: the prize is linear in the discontinuity an attacker can manufacture, and beyond the cap further manipulation is worth exactly nothing.
- **Fact 3 — therefore the attacker maximises ΔA per unit of cost, not ΔA .** This is where the design's threat model and the real one diverge. The design prices the route that buys ΔA of order σ , and does not price the routes that buy ΔA of order 1 at no cost.

8.2 Five routes, one priced

Settlement is immune to on-chain price manipulation, and this is provable rather than assumed: the payoff is an exact even function of the gap, so the sign of any induced move carries zero value, and the index depends on the two prints only through their ratio (check S2, residual $4.22e-15$). That claim is correct and is stronger than the design states. But it covers *one* of five routes into the reference process. The other four are cheaper, and two are not priced anywhere in the design.

Table 21. The five routes into the reference process.

Route	Mechanism	Cost to attacker	Priced in design?
Price manipulation	Move the chain price at the observation timestamp	Prohibitive (settlement ignores chain prices)	Yes — and proved
Definitional	Exploit ambiguity in what the index <i>means</i>	Zero	No
Halt induction	Trigger a trading pause so no valid print arrives	Unknown; no published estimate	Partly (threshold only)
Feed selection	Exploit ties or ordering in the source hierarchy	Zero	No — fixed in §9
Corporate action	Exploit the adjustment mechanism's timing	Zero (it fires on a schedule)	No

8.3 The fallback is a free butterfly

PRINCIPAL SETTLEMENT FINDING

The intuitive fallback rule — refund both legs at half value if no valid print arrives — is a **free long butterfly struck at $c/2$** , written by the protocol at no premium, and its counterparty is the protocol's own hedgers. It transfers up to 0.50 per unit in a single event, requires no capital and no market impact, and it fires on a trigger *correlated with the very risk the instrument exists to hedge*.

The structure is exact and was verified rather than asserted. At $|G| = c/2$ the void is worth exactly nothing — the maximum absolute transfer at that point is 0.0e+00 (check S6) — and the transfer is positive below and negative above, so the void is a claim on the *distance of the gap from $c/2$* . Measured as a function of $|G|$, the quantity $|0.5 - \Pi_L|$ is 0.50 at $A = 0$, 0.00 at $A = c/2$, and 0.50 again at $A \geq c$: a long butterfly struck at $c/2$, written for free (check S7). The per-unit transfer is bounded at 0.50 for every session type (check S16).

The cost is material and quantified. The fallback consumes **29.7 basis points of notional per session** in expectation — **14.3% of the premium the hedger pays** — and the loss is concentrated in exactly the states where the hedge should pay. Critically, the defect is a property of the *rule* rather than of a parameter guess: it holds in all nine combinations of halt band and baseline failure rate tested (checks S10, S17).

8.4 Candidate rules and their costs

Table 22. Five settlement architectures, costed in basis points of notional per session.

Rule	Description	Expected cost	Monotone?	Free option?
R1	Void at 0.50 (the intuitive rule)	29.7 bp	No	Yes
R2	Deferred settlement on the first valid print	0.021 bp	Yes	Zero exactly
R3	Corporate-action-adjusted terminal branch	name-specific	Yes	No
R4	Constant refund with a plausibility band	1.37 bp	No	Reduced
R5	Optimistic challenge window	1.37 bp	Yes	No

The recommended architecture is **R2, deferred settlement**: settle on the first valid print whenever it arrives, with no refund. It removes the free option exactly, passes the monotonicity test that R1 fails, and costs 0.021 bp in expected lockup against R1's 29.7 bp — a factor of roughly 1,400 (checks S8, S13). The earlier analysis understated this: R1 costs only 0.93 bp under a flat void model but 29.7 bp under a threshold model, and the threshold model is the correct one because a halt is not uniform across the gap distribution (check S10).

Two further results support the recommendation. First, the *definitional* route dominates the price route by a factor of at least **3.29×** (check S12) — confirming Fact 3's prediction that the cheap attacks are the definitional ones, not the market-manipulation ones the design defends against. Second, the amended terminal branch is computed on-chain rather than committed as a constant, which matters because a constant 0.50 refund is worth up to 0.50 per unit to whoever can trigger it and 1.3 bp in expectation (check T9). The deployed implementation produces 160165349214785320 for a 0.3% reference gap against a 60-digit reference of 160165279441199520 — a residual of 435634e-12 relative, set by the numerical error-function bound rather than by the model (suite F, check F8).

9. Verification

9.1 Numerical verification

The design was implemented and verified by 105 executable checks across six suites, with the result **94 pass, 10 warn, 1 informational, 0 fail**, in 58.2 seconds. The warnings are findings rather than failures, and are reported as such in §9.4. Selected results:

Table 23. Selected verification results. Residuals are maxima unless stated.

ID	Property	Result
M1	AMM marginal prices sum to exactly one	2.22e-16 over 20,000 pools
M2	Closed form exact; numerical root finder is not	1.80e-15 vs 2.48e-08
M3	Cost and quantity are exact inverses	2.22e-16
M5	Pool invariant preserved by every trade	7.13e-16 over 5,000 trades
M10	Rationalisation is necessary, not cosmetic	naive form max rel. error 2.88
M11	Truncated moment vs quadrature	8.99e-16 over 35 combinations
M12	Truncated moment vs Monte Carlo	within 5 SE, 4M paths each
M14	Strictly increasing in σ (invertibility)	min derivative 4.023e-01 > 0
M15	BELL-IV round-trips	1.16e-13 over 20 combinations
P1	Long + Short = 1 for every gap and leverage	0.000e+00 over 484,121 points
P2	Solvency under randomised stress	0.000e+00 over 400,000 draws
P4	Leverage rule delivers stated saturation	1.00% / 1.02% vs $\alpha = 1\%$
P13	Taxonomy annualisation $\neq$ naive scaling	29.61% vs 25.4%
P15–17	Cap ladder: on-lattice / off-lattice / convex	8.55e-15 / 3.38e-03 / needs signed weights
P22	Volume-weighted median resists manipulation	+1.00% vs VWAP +12.86% at 21% volume
S3	Manipulation prize bound is tight	ratio 1.000000 over 60,000 pairs
K1	All 16 worked-example figures reproduce	at stated precision
K10	Flat-fee annualisation spread	500.6% / 133.7% / 12.17% = 41.1×
X9	Density function correctness	found expWad overflow (see §9.4)

Two verification results deserve emphasis because they are negative results about the *design* rather than the implementation. First, the solvency invariant holds at a residual of exactly zero, not merely within tolerance, across 484,121 grid points and 400,000 randomised draws — Theorem 1 is not merely proved but numerically confirmed at machine exactness. Second, the completeness proposition of §10.1 fails in a specific and design-relevant way: a convex target reproduces to 1.89e-15 with signed weights but only 5.22e-02 with non-negative ones (check P17), which means replicating straddles, butterflies and corridors requires the ability to *write* cap claims, not merely buy them.

9.2 End-to-end verification on a chain fork

The protocol was deployed to a fork of Robinhood Chain at chain ID 4663, L2 block 61,228,000, and exercised through 51 end-to-end checks in six suites. Every core flow passes.

Table 24. Selected end-to-end results on the fork. All are PASS.

ID	Flow	Measured
C1	Sum-to-one on mint	mint 12,345 USDG → equal supplies, solvencyGap == 0
C3	Fee charged once and reachable	10 bp; redeem 10,000 → 9,990; sweep returns gap to 0
C5	Mint-and-swap purchase path	deposit 20,000 → 38,518 Long (18,518 from the swap leg)
C7	Liquidity provision is price-neutral	price moved 0e-12 on add and on remove
C8	Settlement drains collateral to exactly zero	248,833 USDG stranded, recovered exactly; all four balances 0
C9	Cap-and-lever payoff on live prints	gap 1% → 0.19; gap 20% → 1.0; gap -1% → 0.19
D1	Fork identity	chainId 4663; USDG 6 dec; equity tokens 18 dec
D4	Listed ladder matches the reference	196 caps, 0 mismatches, exact integer equality
E2	Precision at small q relative to n	rationalised 500000000000500000 vs naive 500000000000000000
E7	The pool cannot be exhausted	4,000,000 USDG buy vs 250,000 reserve: both legs positive
F1	A session settles exactly once	second resolve reverts; payoff unchanged
F5	Plausibility band refuses feed faults	band 25%; +40% and -90% refused at ingestion
F10	No print refuses to settle	reverts rather than paying a default
N6	Reentrancy guard is load-bearing	nested mintPair refused on both inbound and outbound transfer

Three end-to-end results carry design implications rather than confirming existing behaviour. **D10**: block.number returns the L1 block, not the L2 height — at an L2 pin of 61,228,000 it read 25,962,369, a gap of 35,265,631 blocks. Any logic reasoning about block *number* must say which clock it means; the implementation uses block.timestamp for expiry and is unaffected. **D5**: the live chain supplies a real corporate-action vector, so the terminal branch can be tested against production data rather than a mock. **D8**: the CAL capacity gate operates as a listing condition rather than a claim, at 17.1M committed against a 19M ceiling.

9.3 Failure-mode handling

Twelve documented failure paths were exercised, and the governing principle is stated in one of the checks: a protocol that pays a default when data is missing is a protocol that pays a default. Every path either resolves correctly or refuses explicitly.

- Zero, negative, doubled, NaN, infinite and missing prints are each rejected with a named reason; a 40% move inside the plausibility band still passes (checks F5, F7, T3, T5).
- A missing print causes refusal, not a default payout (F10), and the resolver *raises* rather than voiding — because a void pays 0.50 and would hand out the free option precisely when the feed is broken (T5).
- Settlement before expiry reverts; resolution is permissionless and caller-independent (F11, F12).
- A corporate action forces the terminal branch: the multiplier moving by 1,000 ppm moved the branch from live-print to terminal and changed the payoff from 0.19 to 160165349214785320 (F9).
- Every pre-settlement path is closed after settlement; eight entry points revert (F4).
- A reference token without the corporate-action interface does not brick the session — the probe is a capability test, not a requirement (N1).

9.4 Guards added in this revision, and the deltas that prove them

The ranked gap list asked for something narrower than a passing test suite: a before-and-after. A guard that has never been observed to fire has a known cost and an assumed benefit, and this was the item that changed code

rather than prose. Three of the five guards it named did not exist. They do now, and each is shown firing.

Table 25. The settlement guards, their state before this revision, and the delta test that now exercises each. Every row passes in the G_Guards suite; the full suite is 70 tests across 8 suites with zero failures.

Guard	State before	Delta test	Observed
G9 collateral decimals	Declared in the interface, never called	N3: an 18-decimal collateral, a token with no decimals(), and a 6-decimal control	Reverts CollateralDecimalsUnsupported(18, 6) and CollateralDecimalsUnavailable; the control deploys
G10 issuer pause flag	Absent	G10b: a fresh, in-band print with the issuer paused, then unpaused	Reverts ReferencePaused on preview and resolve; unpausing restores the identical payoff
G10b sequencer uptime	Absent	G10c: an uptime feed reporting down, then up inside and outside the grace period	SequencerDown blocks ingestion and resolution; SequencerGracePeriod(600, 1800) inside the window
G8 multiplier drift	Present, never observed to fire	G8: a -2% headline gap on an ex-date, before and after the multiplier moves	The live branch paid 3.800e17 WAD on the spurious gap; after the drift the terminal corporate-action branch paid 3.422e17 WAD
G3 Tier-1 halt band	Present with a 25% default; the 5% setting untested	G3: the band set to 5%, then a 4% print and a ±6% print	The 4% print is recorded; both ±6% prints revert ImplausibleGap, leaving the print set unchanged

The G8 row is the one the list asked for by name, and it is a genuine before-and-after on an unchanged print set: the same -2% headline gap that a 2% distribution creates on an ex-date settled through the live branch and paid 3.800e17 WAD before the corporate action was recorded, and after the token's multiplier moved from 1e18 to 0.98e18 the identical print set resolved through the terminal corporate-action branch and paid 3.422e17 WAD instead. The ex-date therefore does not print a gap; it changes the branch, which is the design intent.

The G10 row carries a finding worth more than the guard. The first implementation probed every reference token with a paused() staticcall and treated a failed probe as "not paused". Against the pinned fork that turned two passing settlement scenarios red — not with a revert, but with a storage error at the database layer, because the probe drove the real Stock Token into a storage read the RPC provider could not serve before the return value was ever inspected. A staticcall into an unverified selector on a contract with a fallback executes that fallback, so "harmless because we check the return value" holds only if the call returns. The guard is now opt-in per token via setPauseChecked: an unregistered reference is never probed and settles exactly as it did before the guard existed, and a registered one has its pause flag read and honoured. That is a behaviour-preserving default, and it was reached by running the guard against the real chain rather than by reasoning about it.

9.5 Findings that survived

Table 26. Findings, their status, and the required action.

Finding	Severity	Status and action
Collateral decimals unchecked	HIGH	The constructor asserts seven other preconditions but never calls decimals(); UNIT is fixed at 1e6, so an 18-decimal collateral produces a session wrong by 1e12 with no revert, no event and no signal. Not third-party exploitable (the deployer chooses the collateral) but the largest silent assumption in the contract. One-line fix.
Oracle-pause flag unhandled	HIGH	The issuer exposes a pause flag, true while a corporate action is processed, and documents that a paused oracle "may still return a value" — so the flag is advisory and not enforced on-chain. The guard taxonomy runs G1–G9 and none reads it. A staleness check alone is insufficient because the failure mode is a feed frozen but not obviously stale. Requires a tenth guard, paired with — not replacing — the staleness check, plus a sequencer-uptime check.
Multiplier adjustment in the index definition	MEDIUM	Settlement must use the multiplier-adjusted token price, not the share price, or the protocol records a spurious gap on every ex-date (roughly quarterly). The design's reference leg is described as "official external prints," which is ambiguous on exactly this point.
Halt model describes an impossible window	MEDIUM	Price bands and trading pauses operate only during regular trading hours 09:30–16:00 ET, which is precisely <i>not</i> the session BELL prices. The event the model needs is a pause triggered at the open as the bands re-engage against a gapped reference, which is narrower and changes which distribution to condition on.
Regulatory band mismatch	MEDIUM	The design's illustrative 7% halt band matches neither regulatory tier: the correct band for this universe is $\pm 5\%$. Correcting it makes the fallback fire roughly twice as often — 3.03% of sessions have $ G \geq 5\%$ against 1.52% at 7%, about 7.6 events per name per year rather than 3.8. Because the fallback is the settlement defect, the illustrative parameter was flattering the finding.
Mint/burn window excludes the weekend session	MEDIUM	Primary-market creation is unsupported for almost the entire weekend session the protocol prices. Trading is still possible, but any design component assuming weekend primary-market creation assumes something the issuer does not support.
Stale pool biases BELL-IV	Low	A pool frozen at $\sigma = 2.60\%$ reports 2.60% regardless of subsequent volatility. Fixed by provenance stamping with a freshness bound and a named fallback.
Lattice must be specified	Low	Published leverage values depend on an unstated rounding lattice. 0.25% reproduces them; 1% gives 16. On the published 0.25% grid, 68 of 79 caps are not listed markets, so the traded strike set must be the harmonic lattice rather than the grid.

One methodological finding is worth recording because it changed a conclusion. An earlier pass recorded that no Uniswap deployment was reachable on the chain, and treated that as a launch gate. The finding was wrong. Every address probed was an Ethereum mainnet address; a new chain deploys at new addresses, and the canonical set is canonical only on the chain it was deployed to. A zero byte-length at a hardcoded address is evidence about that address and nothing else. Uniswap v2, v3 and v4 are all live on the chain, and the constraint is depth rather than presence — a much smaller problem than having nowhere to trade. **The lesson generalises: a negative result from a probe is only as strong as the address list, and where a finding of the form "X does not exist" is load-bearing, the probe must establish that it searched the space rather than that it tried one address.**

10. Completeness and extensions

10.1 The cap ladder spans the payoff space

One leverage setting gives one payoff shape; a family of settings gives a complete market. This can be stated as a proposition rather than asserted.

Proposition 1 (Spanning). Let $f_c(m) = \min(m/c, 1)$ for $c > 0$. Then for any payoff ψ absolutely continuous on $[0, \infty)$ with $\psi(0) = 0$ and integrable derivative, there exist weights $\{w_j\}$ such that $\sum_j w_j f_{c_j}(m)$ converges uniformly to $\psi(m)$ on every compact set as the lattice $\{c_j\}$ refines.

Proof sketch. Each f_c has derivative $(1/c) \cdot 1_{\{m < c\}}$, a step function. Any weighted sum therefore has a step-function derivative, and step functions are dense in L^1 on compacts. Approximating ψ' arbitrarily well and integrating gives the result. $\square$

Economically, with a dense ladder of caps a trader can synthesise any concave or convex function of the absolute session gap — capped straddles, butterflies, ratio spreads, corridor structures — using only fully-collateralised, no-liquidation primitives. The protocol is not a single instrument; it is a spanning set. Three qualifications emerge from verification. A payoff whose kink sits exactly on a lattice point is reproduced to 8.55e-15, but one whose kink falls between caps is reproduced only to the local lattice spacing, with error falling from 3.74e-02 at a $1.50\times$ lattice to 3.38e-03 as it refines (checks P15, P16). And because a ladder restricted to *buying* claims can only produce concave payoffs, convex targets require signed weights — reproducing to 1.89e-15 with them and only 5.22e-02 without (check P17). The protocol must therefore support writing cap claims, and the collateral, margin and liquidation treatment of a written leg is where the next design question lies. Reassuringly, collateral equals worst-case liability to the last bit on five structurally different ladders, and no path requires a collateral top-up over m in $[0, 2.0]$ — roughly $38\times$ the widest reachable cap (checks T15, T17).

Scope: what spanning does not commit the protocol to. The spanning result is a statement about what the ladder can express, not about what ships at launch. Reproducing a convex payoff requires signed weights on cap claims, which means writing caps, and a written leg reintroduces exactly the collateral, margin and liquidation machinery that the sum-to-one construction exists to eliminate. It is a roadmap item, not an MVP one. The launch ladder therefore supports non-positive weights only: positions that are long some caps and short none, which is every concave payoff in the class, including the capped straddles, butterflies and corridor structures the protocol is actually for. Verification shows the constraint is the binding one rather than a numerical limit — a convex target reproduced with signed weights closes to 1.89e-15, while the same target under the non-negative restriction is left at 5.22e-02, a gap of thirteen orders of magnitude that measures the restriction and not the arithmetic.

10.2 CAL: a derivative on the accuracy of a market

Because a tokenised equity trades continuously while its reference does not, its price is not merely a price — it is a *forecast*. At any moment during the closed session, the token price is the market's estimate of the next official open. That estimate can be evaluated after the fact, and its error can be tokenised.

Fix a designated observation time τ_k inside session k , published in advance and constant across sessions of the same type. Let P_τ be the volume-weighted median token price over a window around τ_k . Define the discovery error

$$D_k = \ln(O_k / P_{\tau_k}) \quad (24)$$

with a CAL-L claim paying $\min(\lambda_D |D_k|, 1)$ and CAL-S its complement, on the same sum-to-one machinery. The payoff is a bet on whether the chain's 24/7 price discovery is well calibrated. Its utility is direct: any protocol that prices risk off the 24/7 token price — a lending market setting collateral haircuts, a market maker quoting off the last print, an agentic trader — is exposed to D , and CAL is the hedge. Its novelty is that the settlement variable is not a price, a rate, or a volatility but the forecast error of a market.

Its risk is that, unlike GAP, the observation is on-chain, so a sufficiently funded actor could move it at the observation timestamp. Two mitigations are quantified. A multi-hour volume-weighted median across venues resists a single large actor: holding 21% of the window volume moves the median by +1.00% while a volume-weighted average price moves +12.86%, and even at 65% of volume the median moves only +6.3% (check P22). And the attack is a capacity condition rather than a property: the break-even attacker position is roughly half the pool depth, so a CAL market may open only where its open interest is small relative to the depth it observes (checks G6, T6). **The design discipline is that GAP and CAL are not equivalent in settlement integrity, and the hierarchy must be explicit: GAP is the flagship and carries the protocol's liquidity and credibility; CAL launches later, on the deepest markets only, with conservative caps.**

The launch gate for CAL. Three conditions must hold together, and none of them is satisfied at the depths the chain shows today. First, CAL opens only on the deepest listed markets. The measured pool depths are \$8.53M for NVDA, \$1.39M for TSLA and \$0.75M for AAPL (§7.6), and at the AAPL figure, moving every venue the observation spans is a low bar rather than a high one — a market whose depth is a rounding error against the flow required to move it should not be opened at all. Second, open interest must stay small relative to the depth observed, because the break-even attacker position is roughly half the pool depth. The attack is a capacity condition, so the gate is a ratio rather than a threshold, and it must be re-checked as depth moves rather than set once at listing. Third, CAL ships only after GAP has accumulated a live record. GAP settles against an external print the protocol does not control; CAL settles against an on-chain observation that the protocol's own venue partly determines. The two are not equivalent in settlement integrity, and launching the weaker one first would spend the credibility the stronger one needs in order to earn.

10.3 The weekend variance ratio as a factor

Equation (16) isolates a factor that exists only because the chain runs 24/7 against a market that does not. A trader can go long the weekend instrument and short three overnight instruments, obtaining pure exposure to WVR with no directional equity exposure and no dependence on the level of volatility. The measurement in §7.2 makes this more interesting rather than less: the design assumed $WVR = 0.88$ and the measurement gives 0.356, and the *sign* of $\sigma_W - \sigma_E$ is not stable across names. A factor whose assumed level was wrong by a factor of 2.5 and whose cross-sectional sign varies with an identifiable mechanism (post-close earnings for mega-cap technology) is a better candidate for a traded factor than one that behaves as modelled.

11. Economics and viability

11.1 Demand

The demand side is the strongest part of the case, because the largest user is not a speculator. It is a participant who is structurally short the gap and has no way to buy protection.

- **Primary — market makers and AMM LPs** quoting token pairs hold inventory across the closed session. They are long the gap with no hedge available. GAP-L is a bounded-cost, no-margin insurance policy sized to inventory.
- **Primary — lending-protocol risk curators.** Venues accepting tokenised equities as collateral face a gap through a liquidation threshold at a moment when the oracle is stale. This is their core tail risk. GAP-L is the hedge and BELL-IV is a direct input to haircut calibration.

- **Secondary — basis and funding desks** who hedge on listed markets during hours are exposed precisely during the hours the listed market is shut.
- **Secondary — volatility sellers.** Buying GAP-S is a bounded short-volatility position on a variable with roughly 252 observations per name per year and a documented empirical distribution — better conditioned than most crypto volatility selling.
- **Secondary — retail, globally.** Most jurisdictions in a 120-country base have no access to US listed options. Maximum loss equals the premium and no margin account is required.
- **Emerging — agentic systems.** Account abstraction and session keys let agents trade on 24/7 prices; an agent needs a machine-readable volatility input and a programmatic hedge.

11.2 Revenue model

Revenue is one line: the term-proportional protocol fee of equation (17). The blended rate across the session mix is **4.106 bp** of open interest (2.397 bp overnight, 8.97 bp weekend, 12.26 bp holiday), which is **10.35% of average open interest per year**, or equivalently **0.879% of hedged inventory per year** once the leverage factor is accounted for. The trading fee of equation (19) is liquidity-provider compensation, not protocol margin: at a 55 bp trading fee the protocol captures 6.9% of the total fee stack, against 29.1% at 10 bp and 3.9% at 100 bp.

Table 27. Revenue scenarios. The design's original figures applied a flat 5 bp; the corrected column uses the blended 4.106 bp.

Scenario	Names	Avg OI / session	Annual notional	Design (5 bp)	Corrected (4.106 bp)
Early	30	\$0.5M	\$3.8B	\$1.89M	\$1.55M
Base	75	\$2.0M	\$37.8B	\$18.9M	\$15.52M
Scaled	150	\$5.0M	\$189B	\$94.5M	\$77.61M

Two structural observations follow. First, the weekend is the revenue engine: it is 20.6% of sessions but **45.1%** of revenue, and weekend plus holiday carry 55.8%. A launch that starts with overnight sessions — the intuitive choice, since they are the most frequent — would target the least valuable part of the calendar. Second, the design's own revenue table is overstated by **21.8%**, because it applies a flat 5 bp where the correct blended rate is 4.106 bp. This is the same defect as the flat protocol fee of \$6.3, appearing in the revenue model rather than the fee schedule.

11.3 The capacity constraint

Revenue scales with open interest held over time, and open interest requires inventory. Inverting the revenue identity gives the required base:

Table 28. Required chain RWA at 25% hedge participation, against available bases. Chain RWA today \$84M; chain TVL \$932.5M; global tokenised-stock market ~\$4B; tokenised-securities market \$35.82B (2026).

Scenario	Revenue	Hedged inventory	Required RWA at 25%	× today's RWA	% of chain TVL
Early	\$1.55M	\$176.3M	\$705.1M	8.4×	75.6%
Base	\$15.52M	\$1,765.1M	\$7,060.3M	84.1×	757%
Scaled	\$77.61M	\$8,826.5M	\$35,306M	420×	3,786%
Break-even, mid opex	\$2.415M	\$274.7M	\$1,098.6M	13.1×	118%

THE BINDING CONSTRAINT

The *Base* case requires \$7.06B of chain RWA at 25% participation — **1.77× the entire global tokenised-stock market**. Even the *Early* case needs \$705M, which is 8.4× today's chain RWA and 76% of the chain's current TVL. Break-even at mid-case opex needs \$1,099M, or 13.1× today's RWA. This is not a defect in the protocol; it is the arithmetic of a derivative whose revenue is a function of the base it hedges.

One unstated variable dominates the result. The calculations above assume all open interest is hedging demand, but if a material share of open interest is speculative — traders taking the other side for its own sake — the required base falls sharply. At 80% speculative share, the *Early* case needs only \$141M of RWA, a factor of 1.7× rather than 8.4×. **The business therefore turns on a number the design never states: the speculative share of open interest.** This is also the variable that determines LP economics, because speculative flow is the flow that balances hedging flow and keeps the pool's net imbalance inside the range where the fee schedule of equation (22) covers the LP's short exposure.

That number is now published as a curve rather than a sentence. Writing the required base as hedged inventory $\times (1 - s) / \text{participation}$, the table below reproduces both of the figures quoted above exactly, at $s = 0$ and $s = 80\%$, which makes it an identity rather than a model. The point of publishing it is that the decision is worth a factor of five in required capital and the answer is not a research question — it is a property of a market that does not yet exist.

Table 29. Required chain RWA in the Early revenue scenario as a function of the speculative share of open interest, at 25% hedge participation. Today's chain RWA is \$84M and chain TVL is \$932.5M. Both published points in §11.3 are reproduced exactly.

Speculative share	Required chain RWA	× today's RWA	% of chain TVL
0%	\$705.2M	8.40×	75.6%
25%	\$528.9M	6.30×	56.7%
40%	\$423.1M	5.04×	45.4%
50%	\$352.6M	4.20×	37.8%
65%	\$246.8M	2.94×	26.5%
80%	\$141.0M	1.68×	15.1%
90%	\$70.5M	0.84×	7.6%

The cheapest way to bound s is not a study of comparable markets, it is the pool itself. The net-flow imbalance is directly observable from the LP's own position from the first session, so a paper-trading or testnet market on the curated launch list measures the distribution that s summarises, at a cost of the order of \$20k against a decision worth a factor of five in required capital. That is the right shape of answer for this variable: a cheap measurement plus a bounded remedy, rather than an open-ended subsidy.

The LP constraint is the harder one. Solving equation (22) for the net-flow imbalance an LP can tolerate at a given turnover gives a range of roughly **3.79%** (time-average trading fees, $1\times$ turnover) to **55.11%** (volatility-weighted fees, $10\times$ turnover). Above that threshold the required subsidy scales with open interest rather than with protocol revenue: an 8% imbalance implies about \$13.7M per year, roughly 91% of open interest; 15% implies \$52.1M, or 347%. **The liability grows at roughly 91% of open interest while the revenue backing it grows at 10.35% — a factor of about nine.** No incentive design repairs a nine-fold mismatch in growth rates, which is why the analysis in §6.4 concludes that the binding constraint on this product is the liquidity-providing side and not the trading side.

11.4 The maker commitment

Lever 2 of the cold-start programme is inventory-backed seeding, and it is the only tolerance in the system that costs the protocol nothing but a commercial agreement. It has been the weakest part of the case because there was no term sheet: no inventory size, no fee split, no quote obligation. The terms below are derived from measured quantities rather than set as negotiating targets, which is what makes them a starting point rather than a wish.

The maker is short a capped straddle. Its expected loss on a session is $\tau \cdot p_L$ of pool value, where τ is the net Long imbalance; its revenue is $\phi \cdot \tau$. Break-even at the measured parameters is exactly the 3.79% reported in §6.4, which means the maker is paid 0.55% of pool value per session — about 139% a year at $1\times$ turnover — for writing that exposure. That number is the whole negotiation, and it is also a competitiveness finding in its own right: 55 basis points per round trip is one to two orders of magnitude above equity-market commissions, and it is what a fully-collateralised, no-margin, bounded-loss structure costs. The fee is not a design choice; it is the price of the risk.

Table 30. Market-maker term sheet. Each term is derived from a measured quantity in this paper rather than set as a target, so the sheet can be priced by a counterparty against the same evidence.

Term	Commitment	Derivation
Committed inventory	\$8.5M per listed name; \$25.5M for the launch set of three	Set at the measured NVDA on-chain depth of \$8.53M across all fee tiers (§7.6), the deepest of the three canonical names. Committing below observed depth makes the pool the marginal venue on every trade it is meant to price
Maximum net exposure	$\pm 3.79\%$ of pool value per session	The break-even imbalance from equation (22) at $1\times$ turnover. Above it the maker is paid less than the option it has sold
Fee split	85% to the maker, 15% to the protocol	The maker carries the entire short-volatility exposure and the protocol carries none of it. The protocol's separate annualised redemption fee is untouched by this split and should be reported as a distinct line rather than netted against it
Quote obligation	Two-sided, within the ramp; maximum spread 2% of mid	The ramp already prices adverse selection into the open, so a width obligation on top of it would double-count. The obligation is to quote, not to quote tight
Uptime	95% of listed sessions, measured monthly	Below this the depth commitment is nominal rather than real
Loss bound	Pool collateral; no margin call, no liquidation	Structural rather than contractual: the pool holds only fully backed tokens, so the worst case is that the retained leg expires worthless. This is the term that makes the book underwritable at all
Subsidy	None at launch; available only on a measured breach of the imbalance band, with a sunset fixed at switch-on	Lever 4 of the cold-start programme, held in reserve. A subsidy without a sunset is a permanent transfer from the protocol to the liquidity side
Termination	30 days' notice after the first 90; immediate on a parameter change	A maker that cannot exit on a parameter change is underwriting governance risk it has not priced

What is missing here is not analysis, it is a counterparty. The inventory number is derived and defensible; whether a maker will commit it at 85% of a 0.55% fee is a question only a maker can answer. That is stated as the open item it is rather than folded into the revenue model, because the whole point of lever 2 is that the protocol's own capital is not the answer.

12. Limitations and open problems

12.1 Unresolvable before launch

- **Net flow imbalance.** The distribution of net Long flow per session is unknown and is a property of a market that does not exist. No amount of research produces it, and the bound on it is a curve rather than a measurement: §11.3 publishes required capital against the speculative share, from $8.40\times$ today's chain RWA at a purely hedged book to $0.84\times$ at a 90% speculative share, reproducing both of the figures the design quotes. This is still the item that makes the viability question genuinely undecidable today, and it is a data question rather than an analysis question. It can be measured only after launch, on a capped, curated list — which is what the staged path proposes — and the measurement is cheap, because the net-flow imbalance is observable from the LP's own position from the first session. What it needs is a running pool, not a study.
- **The cost of inducing a halt.** This protects the void route and is a threshold rather than an estimate, because it requires venue-level market-impact data on the opening cross that is not public. It is genuinely unestimable from documents, and treating it as a threshold rather than inventing a number is the correct treatment.

12.2 Bounded but not resolved

- **The manipulation cost on the price route** now has a measurable input, since reference depth is known. The prize is 0.286 per unit, and moving an \$8.5M pool is materially more expensive than moving a \$0.75M one — so the ranking of the two attack routes is *name-dependent*, which the design treats as a single ranking.
- **The delisting hazard** is conservative, and by a different mechanism than the design states. The design's 5 bp per session is 12.6% per year and is described as conservative by roughly $3\text{--}6\times$ for a US large cap. Measured against pure delisting — bankruptcy or regulatory removal, the event the terminal branch exists for — the rate is 0.2–1.0% a year, or 0.08–0.40 bp per session, which makes the design conservative by $12.6\text{--}63\times$ rather than $3\text{--}6\times$. The $3\text{--}6\times$ figure is reachable only under index removal from all causes, $3\text{--}6\%$ a year, and most of that is merger and acquisition — a corporate action the multiplier path already handles and not a terminal event at all. The design is therefore conservative for the wrong reason, and the terminal branch's value is inflated by a trigger rate borrowed from a different event. The realised count in the 22-name panel is zero, which is consistent with a low rate and says nothing about its level, so the bound stands on published delisting rates rather than on this sample.
- **Per-name calibration.** Resolved for the overnight session, open elsewhere. §7.8 publishes the per-name leverage and premium with bootstrap standard errors, and the bootstrap separates the sessions: $CV(p_L)$ is 5.1–7.5% overnight against 19.1–25.8% on weekend and holiday. The per-name λ spans 10 to 32, so a class-level number is wrong by more than a factor of three at the extremes. Table 14 states the reason as a criterion rather than a judgement: what governs an order statistic is the expected tail count, $n(1 - \alpha)$, which is 19.7 overnight against 4.5 on a weekend, 0.96 on a holiday and 0.34 in an event session. Overnight is publishable, but its interval has to be published with it — AAPL's λ of 22 carries a standard error of 2.0 lattice steps. The weekend's per-name spread is no larger than its own error, so pooling is the only defensible reading. The event session is specified rather than deferred: the shape is pooled and near-Gaussian, the scale stays per-name (§7.10).
- **Sample stationarity.** The window is stated rather than assumed, and the warning is tested rather than cited. §7.9 selects the window by forward error — 42.1% at 126 sessions against 16.9% at 504 for NVDA

overnight — and finds no common drift in the rolling quantile over this sample, with the signs disagreeing across names. The regulatory record's 22.5% year-over-year rise in trading pauses has therefore not yet shown up as a change in the quantile the leverage rule reads. That is an argument for re-testing the window on a schedule, not for believing the warning is wrong: the sample ends at the present, and a regime change in overnight volatility would still move every number.

- **Point-in-time depth.** The pool depths are single reads. They move with the market, and one day's read is not a distribution.

12.3 Structural risks the design cannot mitigate

- **Oracle correctness.** The invariant guarantees solvency, not correctness. A wrong close or open print settles both legs wrongly while the protocol remains perfectly solvent — the invariant protects the system, not the participant. Given that corporate-action-adjusted oracles are an active unsolved problem, this deserves more attention than the design gives it.
- **Platform-level access control.** The chain's terms of service include a reset right allowing the platform to restrict wallet access. This is a genuine centralisation risk for any protocol on the chain and is not mitigable at the protocol layer. It should be disclosed to users rather than glossed over.
- **Regulatory characterisation.** A binary claim settling on whether an equity's open/close ratio crosses a threshold sits close to a regulated event-contract lane — in the United States, the CFTC's event-contract framework under 17 CFR Part 39, where retail access has moved through rulemaking and appellate litigation and is reachable only through a designated contract market with intermediation. Tokenised equities separately raise securities questions under the SEC's regime, and the token wrapper is itself a Jersey tokenised-debt vehicle, so the instrument sits at the intersection of three regimes rather than one. The compliance path is multi-year and capital-intensive, and it is the single largest item on the critical path to a US launch. The design's US exclusion mirrors the token product's own, and jurisdiction gating from day one is the only mitigation available at the protocol layer: the claim is not offered to US persons, eligibility is enforced at the interface rather than by contract, and no parameter is fixed until independent counsel has reviewed the treatment of an index derivative referenced to a tokenised debt security in each jurisdiction where the product would be offered.
- **Venue competition.** The instrument is listed by permission on a venue whose operator also issues the underlying. A first-party closed-session claim would compete with BELL on terms BELL cannot match: no wrapper, no jurisdiction gating, and a distribution channel the protocol does not have. Nothing in the design mitigates this and nothing can. It is stated because it changes what a successful launch looks like: the defensible position is the calibration, the published parameter set and the settlement record rather than the mechanism, since the mechanism is a published theorem that anyone, including the venue, may implement. The practical response is speed and a listing relationship, and it belongs in the launch plan rather than in the design.

12.4 Launch gates

The items in §12.1 and §12.2 are stated as findings. Deployment requires them to become conditions, so each is restated below as a gate with a test that can be failed. The two P0 gates that are not engineering problems — LP cold start and regulatory characterisation — are the ones that decide whether the product reaches mainnet at all; the rest are properties the design already specifies and must be held to.

Table 31. Launch gates. Rank indicates what blocks deployment: P0 gates block mainnet, P1 gates decide competitiveness. Each row is a condition on launch rather than a question for further study.

Gate	Rank	Binding requirement	Closure test
LP cold start	P0	Launch on a capped curated list with committed inventory-backed seeding and volatility-weighted fees; any protocol subsidy requires a measured breach and a sunset fixed at switch-on.	Measured session net-Long imbalance stays inside $\pm 3.79\%$ of pool value at $1\times$ turnover, on every listed name, over the first measured window.
Pricing model	P0	Empirical truncated distribution per name and session is the seed model, calibrated on non-event sessions; the event session carries its own parameters; a fitted fat tail where the sample cannot place the cap; no Gaussian seed.	Met. Per-name leverage and premium are published with bootstrap standard errors (§7.8); a fitted NIG prices the overnight premium within -1.1 to $+2.4\%$ against the Gaussian's $+29$ to $+42\%$ (§7.11); excluding event sessions raises the premium by 4.0 to 16.5% (§7.10).
Settlement route	P0	Deferred settlement on the first valid print (R2). Void at half value (R1) never ships.	Free-option cost ≤ 0.021 bp per session; R1 at 29.7 bp is excluded by test, not by preference.
Settlement guards	P0	Multiplier-adjusted token price in the index definition; a tenth guard for the issuer pause flag beside the staleness check; sequencer-uptime check; $\pm 5\%$ Tier-1 halt band; decimals() asserted at construction. All five are now implemented and delta-tested (§9.4).	Met. Ex-date produces no spurious gap (the live branch paid $3.800e17$ WAD before the drift, the terminal branch $3.422e17$ WAD after); a paused feed is rejected with ReferencePaused and unpausing restores the identical pay-off; SequencerDown blocks both ingestion and resolution; an 18-decimal collateral reverts CollateralDecimalsUnsupported(18, 6); a 5% band admits 4% and refuses $\pm 6\%$.
Regulatory	P0	US excluded; jurisdiction gating enforced at the interface from day one; no parameter fixed before independent counsel review; chain ToS reset right disclosed.	Gating verified before listing; counsel sign-off recorded per jurisdiction; disclosure present in the interface.
Fee guardrail	P1	$\eta_k \leq 0.05 \cdot p_L$, annualised as $12\% \cdot T_k / 8760h$. Keep as specified.	Blended 4.106 bp. The flat-fee alternative annualises $500.6\% / 133.7\% / 12.17\%$ and is rejected on a $41.1\times$ spread.
BELL-IV staleness	P1	Provenance stamp, freshness bound, and a named trailing-realised fallback.	A frozen pool cannot report a stale volatility as current; the fallback engages on the bound.
Premium publisher	P1	Route R5 wired to the off-chain fit: commitment to the parameter set and its inputs before the session, a staleness bound, a bonded dispute resolved by re-running the committed fit, and the trailing-realised fallback while a challenge is open. The 1.37 bp is already carried in the settlement budget, so this adds no cost. The parameters are derived rather than chosen and are collected in Table 19: staleness 12 sessions, publisher bond \$500,000, challenger bond \$50,000, one key per name rotated on the staleness bound, and the §7.9 calibration window for the fallback.	Met. A challenge on a committed parameter set resolves deterministically from the committed inputs, and the pool prices on the fallback for the whole window rather than halting. Without it the pricing trust assumption is unbounded, which is why it is a gate and not a hardening. Implemented in PremiumRegistry.sol and covered by sixteen tests, including the bound and its boundary and both directions of the dispute.
Lattice	P1	Traded strikes are the harmonic ladder, and the rounding lattice is stated in the parameter set.	0.25% reproduces the published leverage values; 1% gives 16; 68 of 79 grid caps are unlistable.
Convex payoffs	P1	Signed weights — writing cap claims — are out of MVP scope.	The launch ladder admits non-positive weights only. $1.89e-15$ with signed weights against $5.22e-02$ without measures the restriction, not the arithmetic.
CAL	P1	Deepest markets only; open interest small relative to observed depth; ships only after GAP has a live record.	The depth ratio is re-checked as depth moves. AAPL at \$0.75M of measured depth fails the ratio test today.
Weekend primary market	P1	No weekend primary-market creation is assumed anywhere in the design.	The mint/burn window (Monday 02:00 to Saturday 02:00 CET) covers almost the whole weekend session; no weekend flow is assumed in any calibration.

Gate	Rank	Binding requirement	Closure test
Stationarity	P1	Rolling recalibration of λ per name and session rather than a static table; overnight window of 252–504 sessions, re-estimated every session; weekend and holiday pooled; point-in-time depth.	Window chosen by forward error (§7.9): 42.1% error at 126 sessions against 16.9% at 504 for NVDA overnight. LULD pauses rose 22.5% year over year, but no common drift appears in the rolling quantile over this sample, so the window is re-tested on a schedule rather than shortened in advance.

13. Conclusion

The closed-session gap is an unusual object: a large, recurring, precisely measurable financial risk that exists only because a venue and its reference market disagree about what time it is. It is manufactured by the structure of a 24/7 chain, it crystallises in exactly the window when listed alternatives are shut, and it is currently warehoused involuntarily and unpriced by market makers and collateralised borrowers.

BELL turns it into an instrument. The mechanism is a pair of complementary claims summing to one, which converts full collateralisation from a risk-management practice into a theorem — no margin, no liquidation, no cascade, and no realised pair of prices at which the protocol owes more than it holds. Verified at a residual of exactly zero over 484,121 grid points and 400,000 randomised draws, and confirmed end-to-end on a fork of the live chain at chain ID 4663.

The work's most useful results are the ones that disagree with the design. Overnight volatility is 36% higher than assumed for the names the product targets, and weekend volatility 13% lower, which inverts the design's own leverage ranking of the two sessions. The weekend variance ratio is 0.356, not 0.88. The Gaussian pricing formula overstates the fair premium by 29 to 42% overnight at the per-name leverage, a variance-matched Student-t repairs only part of that, and a fitted normal-inverse-Gaussian closes it to within 2% — while the family the design named as its fat-tailed reference is rejected on measurement. The fair premium is understated by a further 4 to 16% if event sessions are left in the calibration sample, and the leverage window is settled by forward error rather than by intuition: longer is better on the overnight session, and the weekend session cannot support a per-name window at all. And the intuitive settlement fallback is not a conservative default but a free butterfly written to the protocol's own hedgers, worth 29.7 basis points of notional per session and 14.3% of the premium paid, removable at 0.021 bp by deferring settlement instead of voiding it. Each of these was found by measuring or by proving, not by arguing, and each is stated with the quantity that would falsify it.

Two constraints remain, and neither is a mathematical one. The first is that revenue is a function of open interest held over time, so the base case requires an inventory that is a large multiple of the tokenised-equity market that currently exists — which makes the speculative share of open interest the single variable that decides the business. That variable is now published as a curve rather than a caveat: required chain RWA runs from $8.40\times$ today's for a purely hedged book to $0.84\times$ at a 90% speculative share, and the cheapest way to bound it is a paper-trading market on the curated launch list, at a cost of the order of \$20k against a decision worth a factor of five in required capital. The second is that the binding constraint is not trader demand but liquidity supply: providers in this product class are short volatility by construction, and the liability they absorb grows roughly nine times faster than the revenue that would compensate them. The mechanism is sound, the demand is structural, the calibration data is abundant, and the venue is the only place the product can exist. What remains is a liquidity-side problem and a regulatory one, and both are better addressed with the numbers above in hand than without them.

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Appendix A. Verification inventory

Table A1. Verification suites.

Suite	Scope	Checks
A — Mathematical foundations	AMM algebra, cost function, slippage, truncated moments, BELL-IV inversion, model risk	18
B — Protocol behaviour	Sum-to-one, solvency, leverage rule, fees, LP economics, depth	24
C — Edge cases and failure modes	Degenerate pools, extreme magnitudes, boundary payoffs, invalid inputs	18
D — Design replication	Worked example, tables, figures, arithmetic	11
E — Settlement integrity	Five routes, invariance, free option, candidate rules	17
F — Closure spikes	One executable spike per closed gap	17
Total		105

Table A2. End-to-end suites on the chain fork (chain ID 4663, L2 block 61,228,000).

Suite	Scope	Checks
C	Core flows: mint, redeem, liquidity, purchase paths, payoff, settlement lifecycle	10
D	Fork identity, listing gates, ladders, oracle registry, external venues	10
E	Boundaries, precision, exhaustion, dust	8
F	Failure handling, resolution authority, terminal branches	12
N	Registry gates, reentrancy, notional ceilings, capability probes	6
X	Differential checks against a high-precision reference	14
Total		60

Reproducibility. The numerical verification harness, the measurement scripts and the cached gap sample are available with this paper. The measured statistics in §7 derive from 53,112 close-to-open gaps for 22 US large caps over 2014–2026, pulled from a public historical quote API and cached so the analysis re-runs without re-fetching. Truncated moments are computed on the raw gaps, so the cap is a true truncation rather than a mean of pre-truncated values. On-chain figures are direct reads against the chain at the stated block height. Every table in §7, §9 and §11 is generated from structured data rather than transcribed.

Scope and limitations. This is a protocol design and empirical-calibration paper. It is not investment, legal, tax or accounting advice and has not been reviewed by counsel. The distinction between measured data and illustrative assumption is marked throughout and should be preserved if this document is circulated. Any deployment would require independent legal review of the treatment of index derivatives referenced to tokenised debt securities across the relevant jurisdictions, and independent verification of the empirical gap distributions before any parameter is fixed.